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RESEARCH ARTICLE

iTa-DFiE: An Innovative Tensor Algebra-Based Detection Framework for Incomplete Noninvasive Electroencephalography

NGOC ANH THI NGUYEN¹, QUANG-BANG TAO², HYUNG-JEONG YANG³, (Member, IEEE),
HIEU TRUNG HUYNH⁴, (Member, IEEE), NGUYEN TRAN QUOC VINH¹,
AND DUY KHANH NINH²

¹Faculty of Information Technology, The University of Danang—University of Science and Education, Da Nang 550000, Vietnam

²The University of Danang—University of Science and Technology, Da Nang 550000, Vietnam

³Department of Artificial Intelligence Convergence, Chonnam National University, Gwangju 61186, South Korea

⁴Industrial University of Ho Chi Minh City, Ho Chi Minh City 727900, Vietnam

Corresponding author: Ngoc Anh Thi Nguyen (ntnanh@ued.udn.vn)

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ABSTRACT The paper presents a novel recognition framework for incomplete noninvasive Electroencephalography (EEG) signals relying on the recent advances in tensor algebra, named as An Innovative Tensor Algebra-based Detection Framework for Incomplete Noninvasive Electroencephalography (iTa-DFiE). iTa-DFiE is motivated to improve the diagnostic performance by tackling the major problems shared by a variety of noninvasive EEG-based Brain-Computer Interfaces (BCIs) application is tensorial structured time series with occlusions. The aforementioned challenge setting is solved on two major thrusts, including: 1) tensor completion: discovering hidden patterns and learning their evolving trends to offer missing values imputation via improvement of standard Kalman Filter approach and 2) tensor decomposition: extracting essential hidden information from multi aspects data tensor via extending the most well-known tensor factorization Tucker. The effectiveness and efficiency of the proposed tensor-based framework is proved via successfully improving the pattern classification results on two real-world noninvasive EEG-based motor imagery BCI with diverse corrupted data scenarios, especially in occurrence of consecutive missing observations. Strikingly, iTa-DFiE also outperforms the conventional matrices-based methods and the state-of-the-art tensor techniques in terms of missing reconstructions and feature extraction as well.

INDEX TERMS Brain-computer interfaces, classification, missing imputation, noninvasive electroencephalography, tensor completion, tensor decomposition.

I. INTRODUCTION

BCIs offers a new communication channel with a machine or a computer independently of the user's motor capabilities. Along with the rapid development of BCIs technology, the majority of the research to date has focused on non-invasive scalp electroencephalogram (EEG) modality. According to that, noninvasive EEG is mostly used to develop effective

BCI applications in fields of control and communication and rehabilitation [1], [2], [3], [4], [5], [6], [7], [8].

Originally, EEG data recording preserved multimodal characteristics which often generated from multiple aspects, denoted by a higher-order time series or tensor time series. Indeed, EEG data from each channel can be structured as a time-frequency matrix which indicates temporal and spectral information. Thus, data from multiple electrodes is a sequence of two-dimensional arrays that formed a third-order tensor, spatial $\times$ temporal $\times$ spectral. Intuitively and practically, measurements of EEG time series data are

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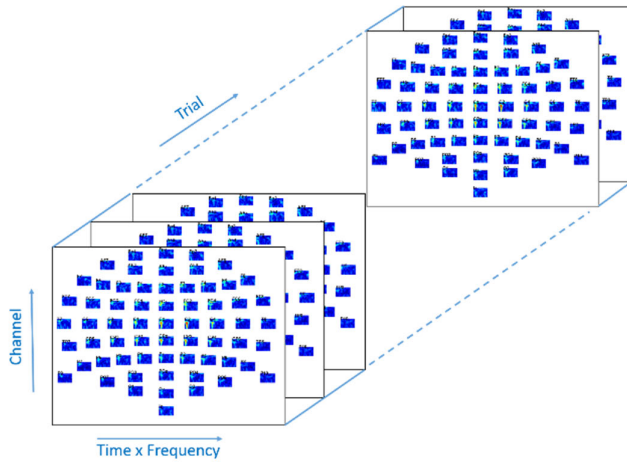


FIGURE 1. Illustration of the four-dimensional tensor comprises of channel, time, frequency, and trial, including 1st, 2nd, 3rd, 4th dimension, respectively.

arranged as a channels $\times$ time-frequency $\times$ measurements or channels $\times$ times-frequencies $\times$ trials. Moreover, with the recent developments in data acquisition technology, further extra modalities can be extended for the dimensions of EEG data, including trials, subjects, dictionaries, conditions, and epochs; therefore, data structure can be formulated to either a fourth-order tensor that is modeled as channels $\times$ time frames $\times$ frequency bin $\times$ trials, or a fifth-order tensor that is modeled as electrodes $\times$ frequencies $\times$ subjects $\times$ conditions $\times$ time frames, or sixth-order tensor as subjects $\times$ frequencies $\times$ trials $\times$ channels $\times$ times $\times$ conditions, illustrated in 4-D tensor in Fig. 1 [9], [10], [11], [12], [13], [14], [15].

For most mathematical models in EEG-based BCI system applying for diagnosis, prediction and treatment, a complete dataset is imperative. Unfortunately, the real-world data from non-invasive EEG signals can be missing during the data acquisition process due to manual mistakes, hardware failure or software failure [16], [17], [18], [19], [20]. More worrisome, the phenomenon of missing observations dramatically causes inaccurate results that would lead to an unreliable diagnosis, or even a reduction in the performance of methodologies. As a result, the corrupted EEG data has posed a great challenge on data analysis and clinical decision-making applications of BCIs systems.

Performing missing data imputation on non-invasive EEG time series data has become a fundamental step before prosing further usage. Unlike most existing studies solves missing observations which have only considered original flat-view matrix or vector models [17], [20], [21], [22], [23], the paper focus here is to deal with occlusion in tensorial structures without destroying inherent multiway nature of the EEG data. Ideally, the paper proposes a comprehensive model for tensor completion of EEG-based BCIs system by decoding the three main properties of brain signals, namely noise, temporal dynamics, and tensor representation simultaneously.

The key idea is to exploit the concept in dynamic graphical model perspective of original Kalman Filter and Tensor algebra. Specifically, the tensor-based Kalman Filter presents its novelty based on improving existing Kalman Filter method whereby maintain limitation on tensor structures via exploring latent variables and evolving behaviors for dealing with imputation, especially for long-interval consecutive missing values.

After presenting missing values imputation successfully, data imputation technique will be evaluated its effectiveness in further developing clinical diagnosis by applying the complete dataset in classification stage. Obviously, to boost an accurate and reliable EEG-based BCIs systems, it is necessary to exhibit feature extraction mechanism as the prerequisite for recognition framework in a principled way [24], [25], [26], [27]. As the result, the paper performs the second contribution via dimensionality reduction for feature extraction and selection to capture compact feature correspond to maximize information about input data that would facilitate the classification.

The fundamental of non-invasive EEG based-BCIs applications is to decode patterns of electric brain activity and convert it into set of actions reflecting the different mental states. Intuitively and practically, the high dimensionality of such multimodal EEG data is a challenge for analysis and interpretation and motivates flexibility approaches to match the intrinsic multilinear pattern properties of EEG data via reducing the dimensionality [28], [29], [30]. This second axis focuses on how to extract compact and meaningful hidden patterns without disruption the original multilinear structures of brain datasets. The tensor-based approach again presents its versatile data analysis tool to retrieve strong discriminative and interpretable features from high-multidimensional EEG time series in general. We herein propose an alternative data-driven approach based on orthogonal Tucker decomposition, which discovering multilinear canonical discriminant constraints that are directly imposed on latent factors of the Tucker approach to reinforce multimodal EEG data via capturing class-label information to different between of classes effectively. The retrieved features are then sent into the classifier model for recognizing brain activities and rehabilitation.

In a nutshell, we proposed an approach of an Innovative Tensor Algebra-based Detection Framework for Incomplete Noninvasive Electroencephalography (iTa-DFiE). It is designed to make novel theoretical and practical contributions in two major thrusts of latent modeling and pattern discovery simultaneously for tensor completion and dimensionality reduction by employing the recent advances in Tensor algebra. iTa-DFiE demonstrates its utility effectively on the real motor imagery BCIs from non-invasive EEG signals via formulating multilinear structure of EEG time series as tensors. The major contributions of this paper can be identified and summarized as follows:

1) **Problem definition:** To the best of our knowledge, we are the first to collectively encode major characteristics on mining incomplete multimodal EEG time series of

brain activity in principled way, including: (a) **Multimodality**: Multiple non-invasive EEG arising from real BCIs applicative refersat are often collected from multi-aspects that naturally constructed as tensor representation (See Fig. 1 for an example), (b) **Missing values**: the prevalence of corrupted signals in EEG poses a big challenge on data-driven decision making in clinical BCIs applications; (c) **temporal dynamics**: this refer to capture simultaneously smoothness and correlation in EEG time series automatically.

2) **Algorithmic framework and Analysis**: The paper proposes iTa-DFiE (the multilinear dynamics systems of Kalman Filter and discriminative tensor Tucker decomposition), a comprehensive framework to deal with all the three aforementioned challenges and investigate an effective and scalability techniques that naturally fits the task of imputation and feature extraction for non-invasive EEG-based BCIs systems recognition.

3) **Empirical Applications**: The experiment is conducted on three openly available motor imagery EEG dataset to validate the robust and superiority of the proposed iTa-DFiE methodology over existing methods in both term of data imputation and hidden patterns discovery.

The remainders of the article progresses are organized in the typical way. Section II briefly introduces the literature review of the EEG time series in terms of data imputation and dimensionality reduction. Afterward, the notations related to tensor are presented in Section III. Section IV defines the problem formally as presents the proposed modeling framework. The empirical study as well as discussions are conducted on realistic data in Section V. Finally, the conclusion is presented in Section VI.

II. RELATED WORK

In this section, previous methodologies are surveyed in two sections to clarify why tensors should be investigated for BCIs systems based on non-invasive EEG time series, including compensating of incomplete entries and methodologies that mainly concern define patterns of electric brain activity to different mental states that would facilitate the classification diagnosis, respectively.

A. MISSING DATA IMPUTATION

One of the straightforward procedures is to ignore or delete the instances that contain EEG missing values [31], [32]. This alternative might lead to the results of data analysis is doubtful because of the loss of significant information, especially if missing observations occur consecutively for a long period of time. As such, instead of discarding the missing observations or presumption of non-invasive EEG data being a completed input, imputation is required to preserve data integrity and boost the reliability of results performance, imputation can be investigated to recover such missing information [21], [22], [23]. Existing studies have been proposed to explore efficient occlusion filling mechanisms for EEG time series which can be classified into categories as following:

1) **Statistical method**: including linear interpolation and splines. These simple interpolation techniques estimate missing values effectively for short gaps by exploiting the smoothness characteristic in time series data [16], [24], [25]. In other words, they estimate missing values based on temporal continuity of time series that expect missing values are close to values in neighboring time ticks and following their moving trends. However, the disadvantage of the interpolation methods is to discard the correlations among multiple channels of EEG over time and they become invalid if the missing gaps are large. Another statistical models can directly apply to address missing values in EEG time series, such as autoregression (AR) and multivariate autoregression (MAR), maximum likelihood-based methods or Bayesian network. However, these works rarely exploit the temporal smoothness and correlations among the observations effectively [33], [34].

2) **Dimensionality reduction and hidden variables models**: the typical missing values imputation strategies related to mechanisms of dimension reduction and latent variable discoveries are Principal Component Analysis (PCA)/ Singular Value Decomposition (SVD) and Missing value SVD (MSVD) which enables to estimate corrupted values in a single sequence by employing the linear correlation across multiple EEG time sequences [35]. However, time evolving of those algorithms are not designed for tracking the ordering of the rows or the column.

3) **Dynamical system**: the state space models can cope with missing values via employing the Expectation-Maximization algorithm to learn the model parameters and detect hidden states at every time ticks of time series. The standard approach of such state space model is Linear Dynamical System (LDS), as known as Kalman Filter or extension of LDS that utilizes their strength and robustness while fitting noise, temporal smoothness, and correlations characteristics of multiple time series for missing recovery [36], [37], [38]. However, LDS is only feasible when the number of missing values is small due to its mechanism of repeatedly constructing the model for estimating the missing values with the current model and updating the model.

4) **Tensor-based methods**: the tensor-based techniques perform their ability for expressing multimodal or multi-aspect data while each of previous surveys face the limitation on this characteristic [39]. The most popular Parallel Factor Analysis (PARAFAC), also known as “Canonical Decomposition (CP)”, and TUCKER can handle missing values in the same way via performing the weight tensor parameters that masks the missing observations via optimization mechanism. Among these two techniques, TUCKER performs more effectively than PARAFAC while recovering missing values in general because TUCKER can discover the hidden factors from tensor data which is not strictly trilinear [40], [41], [42]. However, these methods cannot exploit temporal smoothness or noise capabilities of intrinsic time series. Another alternative method that performs the strengths of high-order time series missing values is Bayesian Probabilistic Tensor

Factorization (BPTF). However, this method has a drawback in terms of the multilinear representation of the latent state [43], [44].

Comparing against all the aforementioned techniques which only cover one or two of the above characteristics of EEG time series, but not all of them. Our proposed iTa-DFiE framework for missing values imputation can be formulating multimodel EEG time series as tensor and adopt tensor completion which encodes noise, temporal smoothness and correlation with multilinear dynamics system in general.

B. DIMENSIONALITY REDUCTION

Practically, the quality of classification performances for non-invasive EEG based-BCIs applications relies strongly on good features extracted from electric brain activity. Therefore, to have efficiency and effectiveness of the classification results, preprocessing analysis methods for determining meaningful discriminative features are considered as the prerequisite for classification mechanism. There are a lot of dimensional reduction methods which are successfully applied on multichannel EEG-based BCI systems, can be categorized into two baseline approaches as following:

1) Vector and Matrix-based method: PCA/SVD and Independent Component Analysis (ICA) are the well-known dimension reduction techniques which exploit linear correlations characteristics across multiple time series EEG-based BCI systems for feature extraction. These linear data models decompose multichannel EEG in spatial and temporal domain to extract significant activities. Another interesting method of linear data model known as nonnegative matrix factorization (NMF) which handles nonnegative data is used recently. In time-frequency domain, the representation of EEG data computed by Fast Fourier Transformation (FFT) with sliding window or wavelet transformation. Another category of feature extraction methods investigates the time-frequency domain on EEG-based BCI are AutoRegression Component (AR), matching filtering (MF). Alternatively, the most successful and widely used feature extraction algorithm for capturing distinctive features is termed the common spatial patterns (CSP). CSP exploits amplitude information of EEG-based BCI in different mental states to detect the spectral discriminators based on calculating discriminative spatial patterns. It projects multichannel EEG into subspace, where the similarities are minimized and the differences between classes are maximized. CSP performs classification accuracy above 90%. The Power spectral density (PSD) method is also a well-known method which exploits amplitude information of multichannel EEG signals to detect spectral discriminative features [45], [46], [47], [48], [49]. The computational models of dimensionality reduction techniques shown above only consider input data as 1D or 2D data structure which may fail in terms of capturing and interpreting the underlying structure of multilinear EEG dataset.

2) Tensor-based method: The following two popular tensor decompositions have been used for most BCI applications, namely PARAFAC and Tucker [50], [51], [52], [53]. In particular, Tucker decomposes tensor input into the core tensor multiplied by a series of factor matrices along with each mode. Thus, to reduce the factor dimensions for feature extraction, the components which are not involved in any major features can be discarded. The family of these tensor decompositions could provide a strategy to the case with a non-equivalent number of optimal components, but the extracted features are not easy to interpret because they cannot preserve one-to-one interaction. Furthermore, these techniques are limited in terms of category information that can be used to boost the classification efficiency since the extract featured can obtain as much information as possible for class discrimination. Recent tensor-based decomposition for discriminative feature extraction are tensor linear Laplacian discriminant analysis (TLLDA) that use contextual-distance-based weighting mechanism for extracting discriminative features from tensor data; local tensor discriminant analysis (LTDA) that extend Linear Discriminant Analysis for discovering the crucial dimensionality for local tensor discriminant.

From the above observations, our framework investigates a new approach of dimensionality reduction for feature extraction step in supervised EEG modeling by proposing an improvement of TUCKER decomposition relies on a discovery of categorical class label in this paper.

III. NOTATION AND PRELIMINARIES

This section briefly introduces the main notations that are used throughout the article. Table 1 summarizes the main symbols along with their definition and mathematical description.

A. TENSORS

A tensor is a multidimensional extension that represents a higher-order generalization of vectors and matrices. In generally, an N th-order tensor is denoted by underlined bold capital letters, $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$. A scalar is zero-way tensor, a vector is a 1-way tensor, and a matrix is a 2-way tensor, denoted by lowercase letter a , bold lowercase letters $\mathbf{a}$, and bold capital letters $\mathbf{A} \in \mathbb{R}^{I \times J}$, respectively. The dimension of $\underline{\mathbf{A}}$ determine the order of a tensor or the number of its mode. For instance, a third-dimensional tensor refers to a third-order tensor or a three-mode tensor [54], [55], [56].

B. VECTORIZATION

The vectorization of a given N th-order tensor $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, denoted by $\text{vec}(\underline{\mathbf{A}}) \in \mathbb{R}^{I_1 I_2 \dots I_N}$, is obtained by iterating elements of tensor $\text{vec}(\underline{\mathbf{A}}) \in \mathbb{R}^{I_1 I_2 \dots I_N}$ into a vector. In particular, the j^{th} element of $\text{vec}(\underline{\mathbf{A}})$ is given by $\text{vec}(\underline{\mathbf{A}})_j = \underline{\mathbf{A}}_{i_1 i_2 \dots i_N}$,

$$\text{Where } j = 1 + \sum_{n=1}^N \prod_{m=1}^{n-1} I_m (i_n - 1) \quad (1)$$

C. MATRICIZATION

The matricization of a tensor $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times J_1 \times J_2 \times \dots \times J_N}$, denoted by $\text{mat}(\underline{\mathbf{A}}) \in \mathbb{R}^{I_1 I_2 \dots I_N \times J_1 J_2 \dots J_N}$, is achieved by converting the tensor structure into a matrix form. In particular, the element of matricization $\text{mat}(\underline{\mathbf{A}})$ is given as following:

$$\text{mat}(\underline{\mathbf{A}})_{kl} = \underline{\mathbf{A}}_{i_1 i_2 \dots i_N j_1 j_2 \dots j_N} \quad (2)$$

$$\text{where } \begin{cases} k = 1 + \sum_{n=1}^N \prod_{m=1}^{n-1} I_m (i_n - 1) \\ l = 1 + \sum_{n=1}^N \prod_{m=1}^{n-1} I_m (j_n - 1) \end{cases}$$

D. TENSOR FACTORIZATION

Given a tensor $\underline{\mathbf{U}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times J_1 \times J_2 \times \dots \times J_N}$, the factorization of tensor $\underline{\mathbf{U}}$ is to decompose it into N factor matrices $\{\underline{\mathbf{U}}^{(n)} \in \mathbb{R}^{I_n \times J_n}\}_{n=1}^N$, then $\underline{\mathbf{U}}_{i_1 i_2 \dots i_N j_1 j_2 \dots j_N} = \prod_{n=1}^N \underline{\mathbf{U}}_{i_n j_n}^{(n)}$. $\underline{\mathbf{U}}$ can be written the Kronecker product of N matrices as following:

$$\text{mat}(\underline{\mathbf{U}}) = \underline{\mathbf{U}}^{(N)} \otimes \underline{\mathbf{U}}^{(N-1)} \otimes \dots \otimes \underline{\mathbf{U}}^{(1)} \quad (3)$$

E. PRODUCT OF CONTRACTED PRODUCT

Given two tensors $\underline{\mathbf{U}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times J_1 \times J_2 \times \dots \times J_N}$ and $\underline{\mathbf{Z}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, the contracted product $\underline{\mathbf{A}} = \underline{\mathbf{U}} \circledast \underline{\mathbf{Z}}$ is given by $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, where:

$$\underline{\mathbf{A}}_{i_1 i_2 \dots i_N} = \sum_{j_1 j_2 \dots j_N} \underline{\mathbf{U}}_{i_1 i_2 \dots i_N j_1 j_2 \dots j_N} \underline{\mathbf{Z}}_{j_1 j_2 \dots j_N} \quad (4)$$

Then, the $\text{vec}(\underline{\mathbf{A}}) = \text{mat}(\underline{\mathbf{U}}) \otimes \text{vec}(\underline{\mathbf{Z}})$. If it is the factorizable with matrices $\{\underline{\mathbf{U}}^{(n)} \in \mathbb{R}^{I_n \times J_n}\}_{n=1}^N$, then:

$$\begin{aligned} \text{vec}(\underline{\mathbf{A}}) &= \text{vec}(\underline{\mathbf{U}} \circledast \underline{\mathbf{Z}}) \\ &= [\underline{\mathbf{U}}^{(M)} \otimes \underline{\mathbf{U}}^{(M-1)} \otimes \dots \otimes \underline{\mathbf{U}}^{(1)}] \text{vec}(\underline{\mathbf{Z}}) \end{aligned} \quad (5)$$

$$\text{where } \text{mat}(\underline{\mathbf{U}}) = \underline{\mathbf{U}}^{(M)} \otimes \underline{\mathbf{U}}^{(M-1)} \otimes \dots \otimes \underline{\mathbf{U}}^{(1)} \quad (6)$$

F. MODE-N TENSOR-MATRIX PRODUCT

Given the tensor $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, the mode- n matrix product between $\underline{\mathbf{A}}$ with the matrix $\underline{\mathbf{U}} \in \mathbb{R}^{J_n \times I_n}$ is denoted by $\underline{\mathbf{Y}} = \underline{\mathbf{A}} \times_n \underline{\mathbf{U}}$ or $\underline{\mathbf{Y}}_{(n)} = \underline{\mathbf{U}} \mathbf{A}_{(n)}$ is the tensor $\underline{\mathbf{Y}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_{n-1} \times J_n \times I_{n+1} \times \dots \times I_N}$. $\underline{\mathbf{Y}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_{n-1} \times J_n \times I_{n+1} \times \dots \times I_N}$ is under elementwise, comprise the following:

$$\begin{aligned} \underline{\mathbf{Y}} &= \underline{\mathbf{A}} \times_n \underline{\mathbf{U}} = (\underline{\mathbf{A}} \times_n \underline{\mathbf{U}})_{i_1 \dots i_{n-1} j_n i_{n+1} \dots i_N} \\ &= \sum_{i_n=1}^{I_n} a_{i_1 \dots i_{n-1} j_n i_{n+1} \dots i_N} u_{j_n i_n} \end{aligned} \quad (7)$$

Alternatively, the multiple mode- n matrix products can be performed in any order, as represented as following:

$$(\underline{\mathbf{A}} \times_n \underline{\mathbf{U}}) \times_m \underline{\mathbf{B}} = (\underline{\mathbf{A}} \times_m \underline{\mathbf{B}}) \times_n \underline{\mathbf{U}} \quad (m \neq n) \quad (8)$$

Briefly, all the possible modes ($n = 1, 2, \dots, N$) of a tensor and a set of matrices $\underline{\mathbf{U}}^{(n)}$ can be expressed by applying multiplication, as given:

$$\underline{\mathbf{A}} \times \{\underline{\mathbf{U}}\} = \underline{\mathbf{A}} \times_1 \underline{\mathbf{U}}^{(1)} \times_2 \underline{\mathbf{U}}^{(2)} \dots \times_N \underline{\mathbf{U}}^{(N)} \quad (9)$$

The multiplication of a tensor, $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ with all but one mode- n , is denoted by the following equation [57], [58]:

$$\underline{\mathbf{A}} \times_{-n} \{\underline{\mathbf{U}}\} = \underline{\mathbf{A}} \times_1 \underline{\mathbf{U}}^{(1)} \times_2 \underline{\mathbf{U}}^{(2)} \dots \times_{n-1} \underline{\mathbf{U}}^{(n-1)} \times_{n+1} \underline{\mathbf{U}}^{(n+1)} \dots \times_N \underline{\mathbf{U}}^{(N)} \quad (10)$$

G. TENSOR NORMAL DISTRIBUTION

Given a tensor $\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$, $\underline{\mathbf{U}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and $\underline{\mathbf{S}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times I_1 \times I_2 \times \dots \times I_N}$, $\underline{\mathbf{A}}$ follows a tensor normal distribution $N(\underline{\mathbf{U}}, \underline{\mathbf{S}})$ with expectation $\underline{\mathbf{U}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ and covariance $\underline{\mathbf{S}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times I_1 \times I_2 \times \dots \times I_N}$ if and only if:

$$\text{vec}(\underline{\mathbf{A}}) \sim N(\text{vec}(\underline{\mathbf{U}}), \text{mat}(\underline{\mathbf{S}})) \quad (11)$$

The normal distribution on tensors can be written as:

$$\underline{\mathbf{A}} \sim N(\underline{\mathbf{U}}, \underline{\mathbf{S}}) \Leftrightarrow \text{vec}(\underline{\mathbf{A}}) \sim N(\text{vec}(\underline{\mathbf{U}}), \text{mat}(\underline{\mathbf{S}})) \quad (12)$$

The tensor Gaussian distribution is defined as a special case of normal distribution on tensors as following distributed with $\underline{\mathbf{C}} \in \mathbb{R}^J$:

$$\begin{aligned} \underline{\mathbf{Z}} &\sim N(\underline{\mathbf{U}}, \underline{\mathbf{G}}) \\ \underline{\mathbf{A}} | \underline{\mathbf{Z}} &\sim N(\underline{\mathbf{C}} \circledast \underline{\mathbf{Z}}, \underline{\mathbf{S}}) \end{aligned} \quad (13)$$

As the result, the marginal distribution of tensor and the posterior distribution given are as following:

The marginal distribution of tensor is defined:

$$\underline{\mathbf{A}} \sim N(\underline{\mathbf{C}} \circledast \underline{\mathbf{U}}, \underline{\mathbf{C}} \circledast \underline{\mathbf{G}} \circledast \underline{\mathbf{C}}^T + \underline{\mathbf{S}}) \quad (14)$$

where $\underline{\mathbf{C}}^T \in \mathbb{R}^J$ and $\underline{\mathbf{C}}_{j_1 \dots j_N i_1 \dots i_N}^T = \underline{\mathbf{C}}_{i_1 \dots i_N j_1 \dots j_N}$, the condition-al distribution of tensor

$$\underline{\mathbf{Z}} | \underline{\mathbf{A}} \sim N(\underline{\mathbf{U}}, \underline{\mathbf{G}}) \quad (15)$$

$$\begin{cases} \underline{\mathbf{U}} = \text{vec}_J^{-1}(\underline{\zeta} + V(\text{vec}(\underline{\mathbf{A}}) - \text{mat}(\underline{\mathbf{C}})\underline{\zeta})) \\ \underline{\mathbf{G}} = \text{mat}_J^{-1}(\tau - V \text{mat}(\underline{\mathbf{C}})\tau) \end{cases} \quad (16)$$

where

$$\begin{cases} \underline{\zeta} = \text{vec}(\underline{\mathbf{U}}) \\ \tau = \text{mat}(\underline{\mathbf{G}}) \\ V = \tau \text{mat}(\underline{\mathbf{C}})^T [\text{mat}(\underline{\mathbf{C}}) \tau \text{mat}(\underline{\mathbf{C}})^T + \text{mat}(\underline{\mathbf{S}})] \end{cases} \quad (17)$$

IV. PROPOSED APPROACH: iTa-DFiE

A. iTa-DFiE FOR MISSING VALUE IMPUTATION

1) PROBLEM DEFINITION

Given an incomplete multilinear EEG data $\underline{\mathbf{X}}$ of finite collection N -th order tensor time series $\underline{\mathbf{X}}_1, \underline{\mathbf{X}}_2, \underline{\mathbf{X}}_3, \dots, \underline{\mathbf{X}}_{T-1}, \underline{\mathbf{X}}_T$ with time duration T , whereby each tensor $\underline{\mathbf{X}}_t = \{x_{i_1 i_2 \dots i_N}\} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ ($1 \leq t \leq T$) shares the same dimension and it denotes as an observation/partially observed tensor at t th time tick.

As such, we can rewrite $\underline{\mathbf{X}} = \{\underline{\mathbf{X}}_1, \underline{\mathbf{X}}_2, \underline{\mathbf{X}}_3, \dots, \underline{\mathbf{X}}_{T-1}, \underline{\mathbf{X}}_T\}$ as $(N+1)$ -th order where $\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$ where I_n ($1 \leq n \leq N$) is the dimensionality of the n th-order and the

TABLE 1. Symbols and definitions.

Symbol	Definition	Mathematical description
$\underline{\mathbf{A}}$	N th-order tensor	$\underline{\mathbf{A}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$
$\mathbf{A}$	Matrix	$\mathbf{A} \in \mathbb{R}^{I \times J}$
A_{ij}	The element at the i th row and the j th column of the matrix $\mathbf{A}$	$\mathbf{A} \in \mathbb{R}^{I \times J}$
$\mathbf{a}$	Vector	$\mathbf{a} \in \mathbb{R}^I$
a	Scalar	$a \in \mathbb{R}$
$\underline{\mathbf{X}}$	A multilinear time series of observations with missing values	$\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$
$\underline{\mathbf{X}}$	Recovered tensor of $\underline{\mathbf{X}}$	$\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$
$\underline{\mathbf{X}}_t$	An observation tensor at the t th time tick	$\underline{\mathbf{X}}_t \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$
$\underline{\mathbf{X}}_o$	The observed values in the tensor sequence $\underline{\mathbf{X}}$	$\underline{\mathbf{X}}_o \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$
$\underline{\mathbf{X}}_m$	Variables for the missing values in the tensor $\underline{\mathbf{X}}$	$\underline{\mathbf{X}}_m \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$
$\underline{\mathbf{W}}$	Missing value indication tensor	$\underline{\mathbf{W}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$
$\underline{\mathbf{Z}}$	A sequence tensor of hidden factors, $\{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$	$\underline{\mathbf{Z}}_t \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$
$\underline{\mathbf{U}}$	The Initial state for hidden variable	$\underline{\mathbf{U}}_0 \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$
$\underline{\mathbf{B}}$	The transition tensor. Factorization of $\underline{\mathbf{B}}$ given:	$\underline{\mathbf{B}}^{(1)} \in \mathbb{R}^{J_1 \times J_1}$, $\underline{\mathbf{B}}^{(2)} \in \mathbb{R}^{J_2 \times J_2}$, ..., $\underline{\mathbf{B}}^{(N)} \in \mathbb{R}^{J_N \times J_N}$
$\underline{\mathbf{D}}$	The projection tensor from hidden state to observation. Factorization of $\underline{\mathbf{D}}$ given	$\underline{\mathbf{D}}^{(1)} \in \mathbb{R}^{I_1 \times J_1}$, $\underline{\mathbf{D}}^{(2)} \in \mathbb{R}^{I_2 \times J_2}$, ..., $\underline{\mathbf{D}}^{(N)} \in \mathbb{R}^{I_N \times J_N}$
$\underline{\mathbf{Q}}$	Transition covariance	$\underline{\mathbf{Q}} \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N \times J_1 \times J_2 \times \dots \times J_N}$
$\underline{\mathbf{Q}}_o$	Initial covariance	$\underline{\mathbf{Q}}_o \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N \times J_1 \times J_2 \times \dots \times J_N}$
$\underline{\mathbf{R}}$	Projection covariance	$\underline{\mathbf{R}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times I_1 \times I_2 \times \dots \times I_N}$
t	A positive integer that increases with time	$1 \leq t \leq T, t \in \mathbb{Z}^+$
T	Duration: length of sequence tensor $\underline{\mathbf{X}}$	$T \in \mathbb{Z}^+$
N	Mode/order of tensor $\underline{\mathbf{X}}$	$N \in \mathbb{Z}^+$
I	The dimension of observation tensor $\underline{\mathbf{X}}$	$I \in (\mathbb{Z}^+)^N$
I_n	The dimension of observation tensor $\underline{\mathbf{X}}$ on mode n	$I_n \in \mathbb{Z}^+$
J	The dimension of the latent tensor	$J \in (\mathbb{Z}^+)^N$
J_n	The dimension of latent tensor on mode n	$J_n \in \mathbb{Z}^+$

last order denotes temporal mode with T dimensions, where T is used to represent the T time ticks of time series.

We define a tensor $\underline{\mathbf{W}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$ as an indicator tensor to indicate the missing values of observations

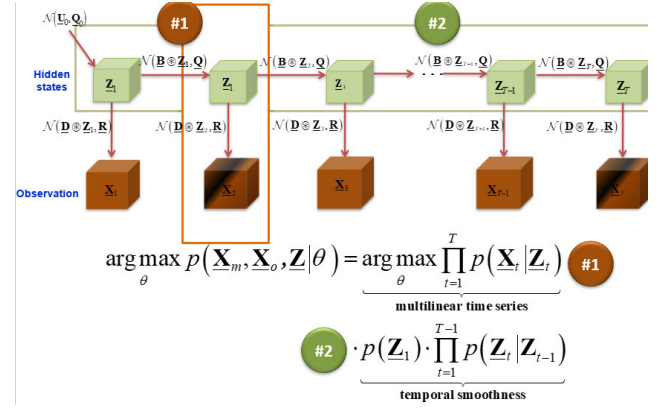


FIGURE 2. Graphical illustration of the probabilistic model of iTa-DFiE for missing value imputation, $\underline{\mathbf{Z}} = \{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \underline{\mathbf{Z}}_3, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$: sequences of latent tensors; $\underline{\mathbf{X}}_1, \underline{\mathbf{X}}_2, \dots, \underline{\mathbf{X}}_{T-1}$: observation tensors; $\underline{\mathbf{X}}_2, \dots, \underline{\mathbf{X}}_T$: partial observation tensors.

of $\underline{\mathbf{X}}$, given by:

$$\underline{\mathbf{W}}_{i_1 i_2 \dots i_{Nt}} = \begin{cases} 0 & \text{if } \underline{\mathbf{X}}_{i_1 i_2 \dots i_{Nt}} \text{ is missing at time } t \\ 1 & \text{if } \underline{\mathbf{X}}_{i_1 i_2 \dots i_{Nt}} \text{ is observed at time } t \end{cases} \quad (18)$$

Accordingly, the model of iTa-DFiE for high-order time series missing value imputation is defined as a network of $\langle \underline{\mathbf{X}}, \underline{\mathbf{X}}_m, \underline{\mathbf{X}}_o, \underline{\mathbf{W}}, \underline{\mathbf{Z}} \rangle$, where $\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$ is a partially observed $(N+1)$ -th order time series tensor, including $\underline{\mathbf{X}}_m \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ indicates missing parts and $\underline{\mathbf{X}}_o \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ presents observed parts; $\underline{\mathbf{W}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times T}$ is the indication tensor that shared the same dimension of $\underline{\mathbf{X}}$; $\underline{\mathbf{Z}} = \{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \underline{\mathbf{Z}}_3, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$ is latent sequence tensor that use to automatically captures the correlation between observation tensors. The complete graphical model of iTa-DFiE for missing imputation is illustrated in Fig. 2.

B. PROPOSED FORMULATION: MODEL SETUP

The proposed model improves the Kalman Filter [59], [60], [61], [62] approach by adding indicator tensor missing and encoding multilinear characteristic of incomplete time series data. Formally, the proposed method first initializes the conditional distribution of the latent factor $\underline{\mathbf{Z}}$ as a tensor normal distribution with the mean $\underline{\mathbf{U}}_0$ and covariance $\underline{\mathbf{Q}}_0$, shown in (19). Consequently, the latent factor $\underline{\mathbf{Z}}_t \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$ is modeled as multilinear Gaussian distribution with multilinear transition tensor $\underline{\mathbf{B}}$ and the covariance $\underline{\mathbf{Q}}$, formulated in (20). The transition tensor $\underline{\mathbf{B}}$ governs dynamics of evolving latent sequence tensor $\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \underline{\mathbf{Z}}_3, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T$, defined as factorization of N factor matrices $\underline{\mathbf{B}}^1, \underline{\mathbf{B}}^2, \dots, \underline{\mathbf{B}}^{N-1}, \underline{\mathbf{B}}^N$, as in (21).

$$\underline{\mathbf{Z}}_1 \sim \mathcal{N}(\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0) \quad (19)$$

$$\underline{\mathbf{Z}}_t | \underline{\mathbf{Z}}_{t-1} \sim \mathcal{N}(\underline{\mathbf{B}} \otimes \underline{\mathbf{Z}}_{t-1}, \underline{\mathbf{Q}}) \quad (20)$$

$$\text{mat}(\underline{\mathbf{B}}) = \underline{\mathbf{B}}^{(N)} \otimes \underline{\mathbf{B}}^{(N-1)} \otimes \dots \otimes \underline{\mathbf{B}}^{(1)} \quad (21)$$

The sequence of observations is generated from series of latent variables $\underline{\mathbf{Z}} = \{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \underline{\mathbf{Z}}_3, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$ with a

projection tensor $\underline{\mathbf{D}}$. The conditional distribution of $\underline{\mathbf{X}}_t$ is formulated as a multilinear normal distribution with the covariance $\underline{\mathbf{R}}$ and the mean is the contracted product of projection tensor and latent factor $\underline{\mathbf{R}}$, defined as equation (22). The projection tensor $\underline{\mathbf{D}}$ also is factorizable and multilinearly transform the latent tensor $\underline{\mathbf{Z}}_t$, shown in (23).

$$\underline{\mathbf{X}}_t | \underline{\mathbf{Z}}_t \sim \mathcal{N}(\underline{\mathbf{D}} \circledast \underline{\mathbf{Z}}_t, \underline{\mathbf{R}}) \quad (22)$$

$$\text{mat}(\underline{\mathbf{D}}) = \underline{\mathbf{D}}^{(N)} \otimes \underline{\mathbf{D}}^{(N-1)} \otimes \dots \otimes \underline{\mathbf{D}}^{(1)} \quad (23)$$

The proposed probabilistic model is built in Fig. 2 in order to estimate: (1) the governing temporal smoothness $\underline{\mathbf{B}}$ and projection tensor $\underline{\mathbf{D}}$ and other parameters; (2) the latent tensor sequence $\underline{\mathbf{Z}}_t = \mathbb{E}[\underline{\mathbf{Z}}_t]$, $t = 1 \dots T$; (3) the expectation $\mathbb{E}[\underline{\mathbf{X}}_m | \underline{\mathbf{X}}_o]$ of miss-ing entries conditioned on the observed values.

Thus, our objective is to estimate the model parameter $\theta = \{\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}, \underline{\mathbf{R}}, \underline{\mathbf{D}}\}$ and find the latent factor $\underline{\mathbf{Z}} = \{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$ that capability maximize the following joint distribution of $\underline{\mathbf{X}}_m, \underline{\mathbf{X}}_o$ and $\underline{\mathbf{Z}}$ is given in (24). The graphical model of proposed method is shown in Fig. 2:

$$\begin{aligned} & \arg \max_{\theta} p(\underline{\mathbf{X}}_m, \underline{\mathbf{X}}_o, \underline{\mathbf{Z}}) \\ &= \arg \max_{\theta} \underbrace{\prod_{t=1}^T p(\underline{\mathbf{X}}_t | \underline{\mathbf{Z}}_t)}_{\text{multilinear time series}} \cdot \underbrace{p(\underline{\mathbf{Z}}_1) \cdot \prod_{t=2}^T p(\underline{\mathbf{Z}}_t | \underline{\mathbf{Z}}_{t-1})}_{\text{dynamics (temporal smoothness)}} \end{aligned} \quad (24)$$

1) OPTIMIZATION FORMULATION: PARAMETER ESTIMATION

The main aim is to fit the iTa-DFiE model by estimating $\theta = \{\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}, \underline{\mathbf{R}}, \underline{\mathbf{D}}\}$. However, it is difficult to maximize Eq. (6) with respect to θ due to the iTa-DFiE model contains latent variables $\underline{\mathbf{Z}} = \{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$. Thus, its local optimal solution is investigated instead following the expectation-maximization (E-M) algorithm. Specifically, the set of parameters θ are first initialized randomly, then E-M steps are performed until satisfying the convergence criterion. In the E step, we estimate the posteriors of the latent variables and its sufficient statistics based on observations and current parameters using mechanism of Kalman Filtering and Kalman smoothing. The model parameters are then updated by maximizing the expectation of the distribution likelihood.

Updating the Latent Factors: E-Step: The tensor normal distribution will facilitate vector and matrix computations rather than working directly with tensor. Thus, with the fix model parameters $\theta = \{\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}, \underline{\mathbf{R}}, \underline{\mathbf{D}}\}$ and $\underline{\mathbf{X}}$ we $\underline{\mathbf{X}}$ first formulate the vectorizations of $\{\underline{\mathbf{X}}, \underline{\mathbf{Z}}\}$ and matricizations of $\{\underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}\}$, as shown in (25), (26), (27).

$$\text{vec}(\underline{\mathbf{Z}}_1) \sim \mathcal{N}(\text{vec}(\underline{\mathbf{U}}_0), \text{mat}(\underline{\mathbf{Q}}_0)) \quad (25)$$

$$\text{vec}(\underline{\mathbf{Z}}_t) | \text{vec}(\underline{\mathbf{Z}}_{t-1}) \sim \mathcal{N}(\text{mat}(\underline{\mathbf{B}}) \text{vec}(\underline{\mathbf{Z}}_{t-1}), \text{mat}(\underline{\mathbf{Q}})) \quad (26)$$

$$\text{vec}(\underline{\mathbf{X}}_t) | \text{vec}(\underline{\mathbf{Z}}_t) \sim \mathcal{N}(\text{mat}(\underline{\mathbf{D}}) \text{vec}(\underline{\mathbf{Z}}_t), \text{mat}(\underline{\mathbf{R}})) \quad (27)$$

Then, our proposed method applies Kalman Filtering (forward) and Kalman smoothing (backward) algorithm to update the expectations and its sufficient statistics of latent factors, as in (28), (29), (30) following:

$$\mathbb{E}[\underline{\mathbf{Z}}_t] = \mathbb{E}[\text{vec}(\underline{\mathbf{Z}}_t)] \quad (28)$$

$$\mathbb{E}[\underline{\mathbf{Z}}_t (\underline{\mathbf{Z}}_t)'] = \mathbb{E}[\text{vec}(\underline{\mathbf{Z}}_t) \text{vec}(\underline{\mathbf{Z}}_t)'] \quad (29)$$

$$\mathbb{E}[\underline{\mathbf{Z}}_t (\underline{\mathbf{Z}}_{t-1})'] = \mathbb{E}[\text{vec}(\underline{\mathbf{Z}}_t) \text{vec}(\underline{\mathbf{Z}}_{t-1})'] \quad (30)$$

Updating the Model Parameters: M-Step:

Updating Non-Multilinear Parameter: in this step, the likelihood of observed data $L(\theta) = p(\underline{\mathbf{X}} | \theta)$ is maximized to result the new non-multilinear parameters estimation. However, maximizing the data likelihood while containing missing values is difficult. Hence, we maximize the expected log-likelihood of the observation sequence, defined with respect to the parameter in $\theta = \{\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}, \underline{\mathbf{R}}, \underline{\mathbf{D}}\}$ in (31).

$$H(\theta) = \mathbb{E}_{\underline{\mathbf{X}}_m, \underline{\mathbf{Z}} | \underline{\mathbf{X}}_o, \underline{\mathbf{w}}} [p(\underline{\mathbf{X}}_o, \underline{\mathbf{X}}_m, \underline{\mathbf{Z}})] \quad (31)$$

With the previous expectations of latent variables are obtained in E-step, taking the derivatives of (31) with respect to the parameter and setting them to zero to estimate new non-multilinear parameters, given results as in (32) and (33), shown at the bottom of the next page.

Updating Multilinear Parameter: The multilinear parameters transition tensor and projection tensor have a specific form which can factorize into N matrices. Therefore, they cannot update in the similar way as non-multilinear parameters due to maintaining different form of likelihood function. Thus, instead of maximizing the likelihood of observed data, we turn to maximizing the expected complete log-likelihood with respect to vector $\mathbf{d}$ via applying the gradient method, shown in [14].

$$\begin{aligned} & L(\mathbf{d}) \\ &= \text{tr} \left\{ \begin{aligned} & \left[\text{mat}(\underline{\mathbf{R}})^{-1} \text{mat}(\underline{\mathbf{D}}) \left[\sum_{t=1}^T \mathbb{E} \left(\text{vec}(\underline{\mathbf{Z}}_t) \text{vec}(\underline{\mathbf{Z}}_t)' \right) \right] \right. \\ & \left. \text{mat}(\underline{\mathbf{D}})' \right] \\ & - 2 \left(\sum_{t=1}^T \text{vec}(\underline{\mathbf{X}}_t) \mathbb{E} \left[\text{vec}(\underline{\mathbf{Z}}_t)' \right] \right) \end{aligned} \right\} \end{aligned} \quad (34)$$

$$\text{Proof: we set } \begin{cases} x_t = \text{vec}(\underline{\mathbf{X}}_t) \\ z_t = \text{vec}(\underline{\mathbf{Z}}_t) \\ \mathbf{R} = \text{mat}(\underline{\mathbf{R}}) \\ \mathbf{D} = \text{mat}(\underline{\mathbf{D}}) \end{cases}$$

$$\begin{aligned} & p(\underline{\mathbf{X}}_1, \underline{\mathbf{Z}}_1, \dots, \underline{\mathbf{X}}_T, \underline{\mathbf{Z}}_T | \theta) \\ &= p(\text{vec}(\underline{\mathbf{X}}_1), \text{vec}(\underline{\mathbf{Z}}_1), \dots, \text{vec}(\underline{\mathbf{X}}_T), \text{vec}(\underline{\mathbf{Z}}_T) | \theta) \\ &= p(x_1, z_1, \dots, x_T, z_T | \text{vec}(\theta)) \\ &= \prod_{t=1}^T p(x_t | z_t) p(z_t) \prod_{t=1}^{T-1} p(z_{t+1} | z_t) \end{aligned}$$

$$\begin{aligned}
&= \prod_{t=1}^N \left\{ \exp \left[(x_t - \mathbf{D}z_t)^T \mathbf{R}^{-1} (x_t - \mathbf{D}z_t) \right] \right. \\
&\quad \left. (2\pi)^{-1} \prod_{n=1}^{J_n/2} |\mathbf{R}|^{-1/2} p(z_1) \prod_{t=1}^{T-1} p(z_{t+1} | z_t) \right\} \\
&\Rightarrow \log p(\mathbf{X}_1, \mathbf{Z}_1, \dots, \mathbf{X}_T, \mathbf{Z}_T | \theta) \\
&= \sum_{t=1}^T \left[(x_t - \mathbf{D}z_t)^T \mathbf{R}^{-1} (x_t - \mathbf{D}z_t) \right] \\
&= \sum_{t=1}^T \text{tr} (x_t - \mathbf{D}z_t)' \mathbf{R}^{-1} (x_t - \mathbf{D}z_t) \\
&= \sum_{t=1}^T \text{tr} \left[(x_t - \mathbf{D}z_t)^T \mathbf{R}^{-1} (x_t - \mathbf{D}z_t) \right] \\
&= \sum_{t=1}^T \text{tr} \left[\mathbf{R}^{-1} (x_t - \mathbf{D}z_t)' (x_t - \mathbf{D}z_t) \right] \\
&= \sum_{t=1}^T \text{tr} \left[\mathbf{R}^{-1} (x_t - \mathbf{D}z_t)' (x_t - \mathbf{D}z_t) \right] \\
&= \sum_{t=1}^T \text{tr} \left[\mathbf{R}^{-1} (x_t x_t' - 2\mathbf{D}z_t z_t' + \mathbf{D}z_t z_t' \mathbf{D}') \right] \\
&= \sum_{t=1}^T \text{tr} \left[\mathbf{R}^{-1} (\mathbf{D}z_t z_t' \mathbf{D}' - 2\mathbf{D}z_t x_t') \right] \\
&= \text{tr} \left[\mathbf{R}^{-1} \mathbf{D} \sum_t z_t z_t' \mathbf{D}' - 2\mathbf{R}^{-1} \mathbf{D} \sum_t z_t x_t' \right] \\
&\Rightarrow \mathbb{E} (\log p(\mathbf{X}_1, \mathbf{Z}_1, \dots, \mathbf{X}_T, \mathbf{Z}_T | \theta)) \\
&= \text{tr} \left[\mathbf{R}^{-1} \mathbf{D} \sum_t \mathbb{E} (z_t z_t') \mathbf{D}' - 2\mathbf{R}^{-1} \mathbf{D} \sum_t \mathbb{E} (z_t x_t') \right]
\end{aligned}$$

where $\mathbf{d}$ is the concatenation of vectorizations of projection matrices $\mathbf{D}^{(n)}$ with $1 \leq n \leq N$, given by:

$$\begin{aligned}
\mathbf{d} &= \left[\text{vec}(\mathbf{D}^{(1)})', \text{vec}(\mathbf{D}^{(2)})', \dots, \text{vec}(\mathbf{D}^{(N-1)})', \right. \\
&\quad \left. \text{vec}(\mathbf{D}^{(N)})' \right] \in \mathbb{R}^{\sum_n I_n J_n}
\end{aligned} \quad (35)$$

For a short form, we set:

$$\begin{aligned}
\omega &= \text{mat}(\mathbf{R})^{-1} \\
\varphi &= \sum_{t=1}^T \mathbb{E} \left[\text{vec}(\mathbf{Z}_t) \text{vec}(\mathbf{Z}_t)' \right] \\
\psi &= \sum_{t=1}^T \text{vec}(\mathbf{X}_t) \mathbb{E} \left[\text{vec}(\mathbf{Z}_t)' \right]
\end{aligned} \quad (36)$$

The equation (14) can be written as following:

$$L(\mathbf{d}) = \text{tr} \left\{ \omega \text{mat}(\mathbf{D}) [\varphi] \text{mat}(\mathbf{D})' - 2\psi \right\} \quad (37)$$

Let l corresponds to $\mathbf{D}_{ij}^{(n)}$ with $1 \leq n \leq N$ and the indicator matrix $\Delta_{ij} \in \mathbb{R}^{I_n \times J_n}$ defined as:

$$c_l = \mathbf{C}_{ij}^{(n)} \text{ with } 1 \leq l \leq \sum_n I_n J_n \quad (38)$$

$$\Delta_{ij} = \begin{cases} 1 & \text{at } (i, j)^{th} \\ 0 & \text{elsewhere} \end{cases} \quad (39)$$

The gradient of the function is given as follows:

$$\begin{aligned}
\nabla L(\mathbf{d}) &= \text{tr} \left\{ \omega \partial_{c_l} \text{mat}(\mathbf{D}) [\varphi] \text{mat}(\mathbf{D})' - \psi' \right\} \\
\text{where } \partial_{c_l} \text{mat}(\mathbf{D}) &= \mathbf{D}^{(N)} \otimes \mathbf{D}^{(N-1)} \otimes \dots \otimes \Delta_{ij} \otimes \dots \otimes \mathbf{D}^{(1)}
\end{aligned} \quad (40)$$

We can compute the trace of the product of two submatrices each with:

$$\nabla L(\mathbf{d})_l = 2 \text{tr} \left\{ \left[\mathbf{D}^{(N)} \otimes \mathbf{D}^{(N-1)} \otimes \dots \otimes \mathbf{D}^{(1)} \right] \Lambda_{ij} \right\} \quad (41)$$

The overall algorithm can be summarized in Algorithm 1.

Similarity, the transition tensor $\mathbf{D}$ can be performed in a same way with $\mathbf{B}$ by replacing the below equation to equation.

$$\begin{aligned}
L(\mathbf{b}) &= \text{tr} \left\{ \begin{aligned} &\text{mat}(\mathbf{Q})^{-1} \text{mat}(\mathbf{B}) \left[\sum_{t=1}^T \mathbb{E} \left(\text{vec}(\mathbf{Z}_t) \text{vec}(\mathbf{Z}_t)' \right) \right] \\ &\text{mat}(\mathbf{B})' \\ &- 2 \left(\sum_{t=1}^{T-1} \mathbb{E} \left[\text{vec}(\mathbf{Z}_{t+1}) \text{vec}(\mathbf{Z}_t)' \right] \right) \end{aligned} \right\}
\end{aligned} \quad (42)$$

Finally, the missing values are easily reconstructed from the fix model parameters and latent variables using Markov

$$\text{vec}(\mathbf{U}_0^{new}) = \mathbb{E} [\text{vec}(\mathbf{Z}_1)] \quad (32)$$

$$\begin{aligned}
\text{mat}(\mathbf{Q}_0^{new}) &= \mathbb{E} \left[\text{vec}(\mathbf{Z}_1) \text{vec}(\mathbf{Z}_1)' \right] - \mathbb{E} [\text{vec}(\mathbf{Z}_1)] \mathbb{E} [\text{vec}(\mathbf{Z}_1)'] \\
\text{mat}(\mathbf{Q}^{new}) &= \frac{1}{T-1} \sum_{t=2}^T \left(\begin{aligned} &\mathbb{E} [\text{vec}(\mathbf{Z}_t) \text{vec}(\mathbf{Z}_t)'] - \text{mat}(\mathbf{B}^{new}) \mathbb{E} [\text{vec}(\mathbf{Z}_{t-1}) \text{vec}(\mathbf{Z}_t)'] \\ &- \mathbb{E} \left[\text{vec}(\mathbf{Z}_t) \text{vec}(\mathbf{Z}_{t-1})' \right] \text{mat}(\mathbf{B}^{new})' + \text{mat}(\mathbf{B}^{new}) \end{aligned} \right) \\
\text{mat}(\mathbf{R}^{new}) &= \frac{1}{T} \sum_{t=1}^T \left(\begin{aligned} &\mathbb{E} [\text{vec}(\mathbf{X}_t) \text{vec}(\mathbf{X}_t)'] - \text{mat}(\mathbf{D}^{new}) \mathbb{E} [\text{vec}(\mathbf{Z}_t)] \text{vec}(\mathbf{X}_t)' \\ &- \text{vec}(\mathbf{X}_t) \mathbb{E} \left[\text{vec}(\mathbf{Z}_t)' \right] \text{mat}(\mathbf{D}^{new})' + \text{mat}(\mathbf{D}^{new}) \end{aligned} \right)
\end{aligned} \quad (33)$$

Algorithm 1 iTa-DFiE for Tensor Completion**Input**

- + The partially observed tensor data: $\underline{\mathbf{X}} = \underline{\mathbf{X}}_o$,
- + Missing indication tensor: $\underline{\mathbf{W}}$,
- + Dimension of latent factors: J .

Output

- + Reconstructed tensor: $\underline{\mathbf{X}}$,
- + Model parameter: $\theta \leftarrow \{\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}, \underline{\mathbf{R}}, \underline{\mathbf{D}}\}$,
- + Latent factors: $\underline{\mathbf{Z}} \leftarrow \{\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T\}$.

Initialization

- + $\underline{\mathbf{X}}_m \leftarrow 0$;
- + $\underline{\mathbf{X}} \leftarrow \underline{\mathbf{X}}_o \cup \underline{\mathbf{X}}_m$;
- + Vectorize $\begin{cases} \underline{\mathbf{X}} \rightarrow \text{vec}(\underline{\mathbf{X}}), \\ \underline{\mathbf{Z}} \rightarrow \text{vec}(\underline{\mathbf{Z}}), \\ \underline{\mathbf{U}}_0 \rightarrow \text{vec}(\underline{\mathbf{U}}_0) \end{cases}$
- + Matricize $\begin{cases} \underline{\mathbf{Q}}_0 \rightarrow \text{mat}(\underline{\mathbf{Q}}_0), \\ \underline{\mathbf{Q}} \rightarrow \text{mat}(\underline{\mathbf{Q}}), \\ \underline{\mathbf{B}} \rightarrow \text{mat}(\underline{\mathbf{B}}), \\ \underline{\mathbf{R}} \rightarrow \text{mat}(\underline{\mathbf{R}}), \\ \underline{\mathbf{D}} \rightarrow \text{mat}(\underline{\mathbf{D}}) \end{cases}$
- + $\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{R}} \leftarrow \text{random values}$;
- + $\theta \leftarrow \{\underline{\mathbf{U}}_0, \underline{\mathbf{Q}}_0, \underline{\mathbf{Q}}, \underline{\mathbf{B}}, \underline{\mathbf{R}}, \underline{\mathbf{D}}\}$.

repeat

- + **Step 1: Estimate the sequence of latent tensors**
(E-step) $\underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T = \text{E}[\underline{\mathbf{Z}} | \underline{\mathbf{X}}, \theta]$ via Kalman Filtering and Kalman smoothing to obtain expectation of its sufficient statistics:

$$\text{E}[\text{vec}(\underline{\mathbf{Z}}_t)], \text{E}[\text{vec}(\underline{\mathbf{Z}}_t) \text{vec}(\underline{\mathbf{Z}}_t)'], \text{E}[\text{vec}(\underline{\mathbf{Z}}_t) \text{vec}(\underline{\mathbf{Z}}_{t-1})']$$

- + **Step 2: Estimate the new model parameters (M-step)**

- **Updating non-multilinear parameters:**

- $\text{vec}(\underline{\mathbf{U}}_0), \text{mat}(\underline{\mathbf{Q}}_0), \text{mat}(\underline{\mathbf{Q}}), \text{mat}(\underline{\mathbf{R}})$
- by maximizing the likelihood:

$$\begin{aligned} L(\theta | \underline{\mathbf{Z}}_1, \underline{\mathbf{Z}}_2, \dots, \underline{\mathbf{Z}}_{T-1}, \underline{\mathbf{Z}}_T, \underline{\mathbf{X}}) \\ = L\left(\text{vec}(\theta) | \text{vec}(\underline{\mathbf{Z}}_1), \text{vec}(\underline{\mathbf{Z}}_2), \dots, \text{vec}(\underline{\mathbf{Z}}_{T-1}), \text{vec}(\underline{\mathbf{Z}}_T), \underline{\mathbf{X}}\right) \end{aligned}$$

- **Updating non-multilinear operators:**

$$\underline{\mathbf{B}} \mapsto \text{mat}(\underline{\mathbf{B}}), \underline{\mathbf{D}} \mapsto \text{mat}(\underline{\mathbf{D}})$$

- + **Step 3: Update the missing value along time mode.**

forall $I_1, I_2, \dots, I_N$

$$\text{vec}(\underline{\mathbf{X}}_t^{\text{new}}) = \text{mat}(\underline{\mathbf{D}}) \text{E}[\text{vec}(\underline{\mathbf{Z}}_t)]$$

until convergence

property, as shown in the graphical model in Fig. 2 and in (43). In vectorization space, the recovered time series can be obtained as in (44). In other word, the missing values of $\underline{\mathbf{X}}_t$ can be inferred from the corresponding entries of $\text{vec}(\underline{\mathbf{X}}_t)$, as given:

$$\text{E}[\underline{\mathbf{X}}_m | \underline{\mathbf{X}}_o, \underline{\mathbf{Z}}; \theta] = \underline{\mathbf{D}} \cdot \text{E}[\underline{\mathbf{Z}}]_{\{i_1, i_2, \dots, i_N\}}, (\{i_1, \dots, i_N\} \in \underline{\mathbf{W}}) \quad (43)$$

$$\text{vec}(\underline{\mathbf{X}}) = \text{mat}(\underline{\mathbf{D}}) \text{E}[\text{vec}(\underline{\mathbf{Z}}_t)] \quad (44)$$

C. iTa-DFiE FOR FEATURE EXTRACTION

This section presents the formulation for the problem of feature extraction as well as its learning based on extending the Tucker decomposition method.

1) PROPOSED FORMULATION: MODEL SETUP

Given a concatenated tensor of K training samples that formed as $(N+1)$ -th order tensor, denoted by $\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times K}$, $k = 1, 2, \dots, K$. Each training data is presented as N -th order tensor of $\underline{\mathbf{X}}^{(k)} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ that belongs to the class with the class label c_k .

In the general feature extraction mechanism of Tucker, tensor $\underline{\mathbf{X}}$ is formulated by a core tensor $\underline{\mathbf{G}} \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$ and a set of basic factor matrix $\underline{\mathbf{A}}^{(n)} \in \mathbb{R}^{I_n \times J_n}$ that represents the reduced features of $\underline{\mathbf{X}}$ and feature space respectively [58], shown as in (45).

$$\underline{\mathbf{X}} = \underline{\mathbf{G}} \times_1 \underline{\mathbf{A}}^{(1)} \times_2 \underline{\mathbf{A}}^{(2)} \times \dots \times_N \underline{\mathbf{A}}^{(N)} \quad (45)$$

The key idea of the proposed approach is to employ Tucker decomposition for dimension-reducing decomposition based on comprising multilinear discriminative analysis. In other words, the improvement of the proposed method lies in exploiting the category information from between the classes. Consequently, the proposed algorithm looks for the set of the discriminative basic factors in order to obtain the compression of core tensor $\underline{\mathbf{G}} \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$ from the same classes are as similar as possible and vice versa.

2) PROPOSED OPTIMIZATION ALGORITHM

To obtain the discriminative factors, the proposed method defines the cost function optimization in (46).

$$\begin{aligned} \psi \\ = \arg \max_{\underline{\mathbf{A}}^{(1)}, \dots, \underline{\mathbf{A}}^{(N)}} \\ \left(\sum_{c=1}^C K_c \left\| \underline{\mathbf{G}}^{(c)} - \underline{\overline{\mathbf{G}}}^{(c)} \right\|_F^2 / \sum_{k=1}^K K_c \left\| \underline{\mathbf{G}}^{(k)} - \underline{\overline{\mathbf{G}}}^{(c_k)} \right\|_F^2 \right) \end{aligned} \quad (46)$$

where $\underline{\mathbf{G}}^{(c)}$: mean tensor of the c -class,

$c = 1, \dots, C$: class labels (class categories)

c_k : class label that k -th training sample $\underline{\mathbf{X}}^{(k)}$ belongs to

$\underline{\overline{\mathbf{G}}}$: mean tensor of all training features

The learning rule of algorithm here is to estimate unknown basic factor $\underline{\mathbf{A}}^{(n)}$ by assuming all factor matrix $\underline{\mathbf{A}}^{(1)}, \underline{\mathbf{A}}^{(2)}, \dots, \underline{\mathbf{A}}^{(n-1)}, \underline{\mathbf{A}}^{(n+1)}, \dots, \underline{\mathbf{A}}^{(N)}$ are fixed. The category information is exploited via an optimization of (46) whereby both between-class and within-class scatter matrices are calculated.

The between-class scatter can be expressed as trace of the between-class scatter matrix, as following:

$$\begin{aligned} \sum_{c=1}^C K_c \left\| \underline{\mathbf{G}}^{(c)} - \underline{\overline{\mathbf{G}}}^{(c)} \right\|_F^2 \\ = \sum_{c=1}^C K_c \left\| \left(\underline{\mathbf{X}}^{(c)} \times_{-n} \{\underline{\mathbf{A}}^T\} \right) \times_n \underline{\mathbf{A}}^{(n)T} - \left(\underline{\overline{\mathbf{X}}} \times_{-n} \{\underline{\mathbf{A}}^T\} \right) \times_n \underline{\mathbf{A}}^{(n)T} \right\|_F^2 \\ = \sum_{c=1}^C K_c \left\| \left(\underline{\mathbf{X}}^{(c)} - \underline{\overline{\mathbf{X}}} \right) \times_{-n} \{\underline{\mathbf{A}}^T\} \times_n \underline{\mathbf{A}}^{(n)T} \right\|_F^2 \end{aligned}$$

$$\begin{aligned}
&= \sum_{c=1}^C \left\| \left(\widehat{\mathbf{P}}^{(c)-n} \times_n \mathbf{A}^{(n)T} \right) \right\|_F^2 \\
&= \sum_{c=1}^C \text{tr} \left[\mathbf{A}^{(n)T} \left\langle \widehat{\mathbf{P}}^{(c)-n}, \widehat{\mathbf{P}}^{(c)-n} \right\rangle_{-n} \mathbf{A}^{(n)} \right] \\
&= \text{tr} \left[\mathbf{A}^{(n)T} \left(\sum_{c=1}^C \left\langle \widehat{\mathbf{P}}^{(c)-n}, \widehat{\mathbf{P}}^{(c)-n} \right\rangle_{-n} \mathbf{A}^{(n)} \right) \right] \\
&= \text{tr} \left[\mathbf{A}^{(n)T} \mathbf{S}_b^{-n} \mathbf{A}^{(n)} \right] \quad (47)
\end{aligned}$$

where $\widehat{\mathbf{P}}^{(c)-n}$ is the c -part of tensor $\widehat{\mathbf{P}}^{-n}$, $\mathbf{S}_b^{-n}$ is the between-class scatter matrix.

$$\begin{aligned}
\widehat{\mathbf{P}}^{(c)-n} &= \widehat{\mathbf{X}}^{(c)} \times_{-n} \{\mathbf{A}^T\} \\
\widehat{\mathbf{P}}_c^{-n} &= \widehat{\mathbf{P}}^{-n} \\
\widehat{\mathbf{X}}^{(c)} &= \sqrt{K_c} \left(\widehat{\mathbf{X}}^{(c)} - \overline{\widehat{\mathbf{X}}} \right) \\
\widehat{\mathbf{P}}^{-n} &= \widehat{\mathbf{X}} \times_{-(n.N+1)} \{\mathbf{A}^T\} \\
\mathbf{S}_b^{-n} &= \sum_{c=1}^C \left\langle \widehat{\mathbf{P}}^{(c)-n}, \widehat{\mathbf{P}}^{(c)-n} \right\rangle_{-n} = \left\langle \widehat{\mathbf{P}}^{-n}, \widehat{\mathbf{P}}^{-n} \right\rangle_{-n}
\end{aligned}$$

Similarity, the within-class scatter can be expressed as trace of the within-class scatter matrix, as following:

$$\begin{aligned}
&\sum_{k=1}^K K_c \left\| \mathbf{G}^{(k)} - \overline{\mathbf{G}}^{(c_k)} \right\|_F^2 \\
&= \sum_{k=1}^K \left\| \left(\mathbf{X}^{(k)} \times_{-n} \{\mathbf{A}^T\} \right) \times_n \mathbf{A}^{(n)T} - \left(\overline{\mathbf{X}}^{(c_k)} \times_{-n} \{\mathbf{A}^T\} \right) \times_n \mathbf{A}^{(n)T} \right\|_F^2 \\
&= \sum_{k=1}^K \left\| \left(\mathbf{X}^{(k)} - \overline{\mathbf{X}}^{(c_k)} \right) \times_{-n} \{\mathbf{A}^T\} \times_n \mathbf{A}^{(n)T} \right\|_F^2 \\
&= \sum_{k=1}^K \left\| \widehat{\mathbf{P}}^{(k)-n} \times_n \mathbf{A}^{(n)T} \right\|_F^2 \\
&= \sum_{k=1}^K \text{tr} \left[\mathbf{A}^{(n)T} \left\langle \widehat{\mathbf{P}}^{(k)-n}, \widehat{\mathbf{P}}^{(k)-n} \right\rangle_{-n} \mathbf{A}^{(n)} \right] \\
&= \text{tr} \left[\mathbf{A}^{(n)T} \left(\sum_{k=1}^K \left\langle \widehat{\mathbf{P}}^{(k)-n}, \widehat{\mathbf{P}}^{(k)-n} \right\rangle_{-n} \right) \mathbf{A}^{(n)} \right] \\
&= \text{tr} \left[\mathbf{A}^{(n)T} \mathbf{S}_w^{-n} \mathbf{A}^{(n)} \right] \quad (48)
\end{aligned}$$

where $\widehat{\mathbf{P}}^{(k)-n}$ is the k -part of tensor $\widehat{\mathbf{P}}^{-n}$, $\mathbf{S}_w^{-n}$ is the between-class scatter matrix.

$$\begin{aligned}
\mathbf{P}^{(k)-n} &= \mathbf{X}^{(k)} \times_{-n} \{\mathbf{A}^T\} \\
\mathbf{P}_k^{-n} &= \mathbf{P}^{-n} \\
\mathbf{X}^{(k)} &= \mathbf{X}^{(k)} - \overline{\mathbf{X}}^{(c_k)} \\
\mathbf{P}^{-n} &= \mathbf{X} \times_{-(n.N+1)} \{\mathbf{A}^T\} \\
\mathbf{S}_w^{-n} &= \sum_{k=1}^K \left\langle \mathbf{P}^{(k)-n}, \mathbf{P}^{(k)-n} \right\rangle_{-n} = \left\langle \mathbf{P}^{-n}, \mathbf{P}^{-n} \right\rangle_{-n}
\end{aligned}$$

Hence, the discriminant basic factor $\mathbf{A}^{(n)}$ from the (46) can be achieved via maximizing the trace ratio of the between-class and within-class scatter matrix, as following:

$$\begin{aligned}
\psi &= \arg \max_{\mathbf{A}^{(n)}} \frac{\text{tr} [\mathbf{A}^{(n)T} \mathbf{S}_b^{-n} \mathbf{A}^{(n)}]}{\text{tr} [\mathbf{A}^{(n)T} \mathbf{S}_w^{-n} \mathbf{A}^{(n)}]} \\
&= \arg \max_{\mathbf{A}^{(n)}} \text{tr} [\mathbf{A}^{(n)T} \mathbf{S}_w^{-1} \mathbf{S}_b \mathbf{A}^{(n)}] \quad (49)
\end{aligned}$$

The overall algorithm can be summarized in Algorithm 2.

With the given basic factors $\mathbf{A}^{(n)} \in \mathbb{R}^{I_n \times J_n}$, $n = 1, 2, \dots, N$ the features of testing tensor $\mathbf{X}^{(t)} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$ are extracted via finding the core tensor $\mathbf{G}^{(t)} \in \mathbb{R}^{J_1 \times J_2 \times \dots \times J_N}$ in (50). The basic factor $\mathbf{A}^{(n)}$ is orthogonal, the core feature of $\mathbf{G}^{(t)}$ can be easily obtained by the tensor-matrix products in (52).

$$\mathbf{X}^{(t)} = \mathbf{G}^{(t)} \times_1 \mathbf{A}^{(1)} \times_2 \mathbf{A}^{(2)} \times \dots \times_N \mathbf{A}^{(N)} \quad (50)$$

$$\mathbf{G}^{(t)} = \mathbf{X}^{(t)} \times_1 \mathbf{A}^{(1)\dagger} \times_2 \mathbf{A}^{(2)\dagger} \times \dots \times_N \mathbf{A}^{(N)\dagger} \quad (51)$$

$$\mathbf{G}^{(t)} = \mathbf{X}^{(t)} \times_1 \mathbf{A}^{(1)T} \times_2 \mathbf{A}^{(2)T} \times \dots \times_N \mathbf{A}^{(N)T} \quad (52)$$

As summary, the proposed approach is motivated by exploiting the class label information of multilinear EEG time series for feature extraction, the pseudocode of this problem is given in Algorithm 2. The retrieved features are used in the classification step.

V. EXPERIMENTAL RESULT

A. EXPERIMENT SETUP

1) DATA DESCRIPTION

We conducted experimental results on three publicly available data sets of high-dimensional EEG. We formulate the same tensor structure “channels x times x trials” for each real dataset which is a typical tensor algebra data structure of multidimensional EEG-based BCI dataset. Specifically, channels indicate the number of electrodes that use to record signals; times denote the number of samples in each imagination task and trials are the number of motor imagery performed in this session, as follows:

(Gr) BCI motor imagery competition from Graz University of Technology: The dataset contains 140 trials that were

Algorithm 2 Algorithm 2: iTa-DFiE for Tensor Decomposition**Input**+ Given an k -th training sample tensor:

$$\mathbf{X}^{(k)} = \{x_{i_1 i_2 \dots i_N}\}^{(k)} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N}$$

+ A concatenated tensor of K training samples:

$$\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times \dots \times I_N \times K}, k = 1, 2, \dots, K$$

Output+ N discriminative factor matrices:

$$\underline{\mathbf{A}}^{(n)} \in \mathbb{R}^{I_n \times J_n}, n = 1, 2, \dots, N$$

+ Training core feature tensor: $\underline{\mathbf{G}} \in \mathbb{R}^{I_1 \times J_2 \times \dots \times J_N \times K}$ **Initialization**All factors $\underline{\mathbf{A}}^{(n)} \leftarrow$ random values**repeat****for** $n = 1$ to N **do**// assuming all factor matrices are fixed except in mode- n $\underline{\mathbf{A}}^{(n)}$

// calculating within-class scatter matrix

$$\begin{cases} \underline{\mathbf{F}} = \underline{\mathbf{P}} \times_{-(n, N+1)} \{\mathbf{A}^T\} \\ \underline{\mathbf{S}}_w^{-n} = \left(\underline{\mathbf{P}}^{-n}, \underline{\mathbf{P}}^{-n} \right)_{-n} \end{cases}$$

// calculating between-class scatter matrix:

$$\begin{cases} \underline{\mathbf{P}} = \underline{\mathbf{P}} \times_{-(n, N+1)} \{\mathbf{A}^T\} \\ \underline{\mathbf{S}}_b^{-n} = \left(\underline{\mathbf{P}}^{-n}, \underline{\mathbf{P}}^{-n} \right)_{-n} \end{cases}$$

// Finding the discriminative factors by maximizing:

$$\begin{aligned} \psi &= \arg \max_{\underline{\mathbf{A}}^{(n)}} \frac{\text{tr} [\underline{\mathbf{A}}^{(n)T} \underline{\mathbf{S}}_b^{-n} \underline{\mathbf{A}}^{(n)}]}{\text{tr} [\underline{\mathbf{A}}^{(n)T} \underline{\mathbf{S}}_w^{-n} \underline{\mathbf{A}}^{(n)}]} \\ &= \arg \max_{\underline{\mathbf{A}}^{(n)}} \text{tr} [\underline{\mathbf{A}}^{(n)T} \underline{\mathbf{S}}_w^{-1} \underline{\mathbf{S}}_b \underline{\mathbf{A}}^{(n)}] \end{aligned}$$

end**until convergence**

recorded from 25 years of female subjects. Each trial lasts 9s in feedback session and collected by three channels of C3, CZ and C4 bipolar EEG channels with sampling rate 128 Hz. This task of data is to control a feedback of two-classes motor imagery of her hand movements: imagery left or right-hand movements. The data size in the form of tensor structure is $3 \times 1152 \times 280$.

(Fr) BCI III motor imagery competition benchmark is published by Fraunhofer FIRST, Intelligent Data Analysis Group and Campus Benjamin Franklin of the Charité - University Medicine Berlin, Department of Neurology, Neurophysics Group. The data set was recorded from 118 EEG channels and markers that indicate the time points of 280 cues for each of the 5 healthy subjects (aa, al, av, aw, ay). All signals were band-pass filtered between 0.05 and 200 Hz. In this paper, subject aw (BCI-aw) and al (BCI-al) are selected for our implementation and their size in tensor form are $118 \times 350 \times 280$.

(Ri) BCI EEG motor imagery dataset is recoded by Riken brain science institute: The EEG-based BCI data were collected from several healthy subjects that consisted of two motor imagery tasks, namely the imagination of movement of the left hand and right hand. In this paper, there are two

subjects that were selected to do in our experiments, namely Subject A (SubA) and Subject B (SubB). Data “SubA” consists of 130 trials of BCI EEG motor imagery which were recorded in a duration of 3 seconds with a sampling frequency of 256 Hz over 6 channels by a gTec amplifier, its data size in tensor form is $6 \times 768 \times 130$. EEG signals first go through a bandpass filter (8-30 Hz). Data “SubB” consists of 162 trials of BCI EEG motor imagery that were recorded in a duration of 4 seconds at a sampling frequency of 250 Hz over 6 channels by a Neuroscan amplifier and its data size in tensor form is $6 \times 1000 \times 162$.

2) BASELINE METHODS FOR COMPARISON

We compare the effectiveness of the proposed framework iTa-DFiE against the following baseline models in both tasks of missing imputation and feature extraction, as given:

Missing value imputation: linear interpolation, spline interpolation, Missing value Singular Decomposition Values (MSVD) and original Kalman Filter (KF).

Feature extraction: (1) the matrix-based factorization models including: Principal Component Analysis (PCA), Common Spatial Pattern (CSP); (2) the tensor-based factorization models including: Tensor linear Laplacian discriminant analysis (TLLDA); Local tensor discriminant analysis (LTDA) and the original Tucker decomposition.

3) EXPERIMENT SETTING

Practically, EEG-based BCI systems are multidimensional time series containing various missing patterns as well as missing ratios. Thus, we mimic two real missing scenarios, including the element-like and consecutive element-like random missing in the numerical case studies with the missing rate of original tensor raising from 10% to 50% (step 10%). In detail, the scenario of element-like missing (EM) values indicates that individual data points are lost from the original tensor at random. While the consecutive element-like random missing (CEM) denotes that data points are missing continuously during a few time periods. This problem is more common and realistic in EEG missing patterns. To experiment on synthetic consecutive missing values, we implement by fixing 10% of missing rate and run on different missing length. The intuitive visualization of two cases is shown in Fig. 3.

4) EVALUATION MEASURES

Reconstruction Error: To evaluate the quality of missing value recovery, the root mean squared error (RMSE) criteria is used to measure reconstruction performance. For a matrix-based imputation, we define the RMSE between the original data $\mathbf{X}$ and the estimated missing values data $\hat{\mathbf{X}}$, as formally:

$$RMSE = \sqrt{\frac{\sum_{it} (1 - \mathbf{W}_{it}) (\mathbf{X}_{it} - \hat{\mathbf{X}}_{it})^2}{\sum_{it} (1 - \mathbf{W}_{it})}} \quad (53)$$

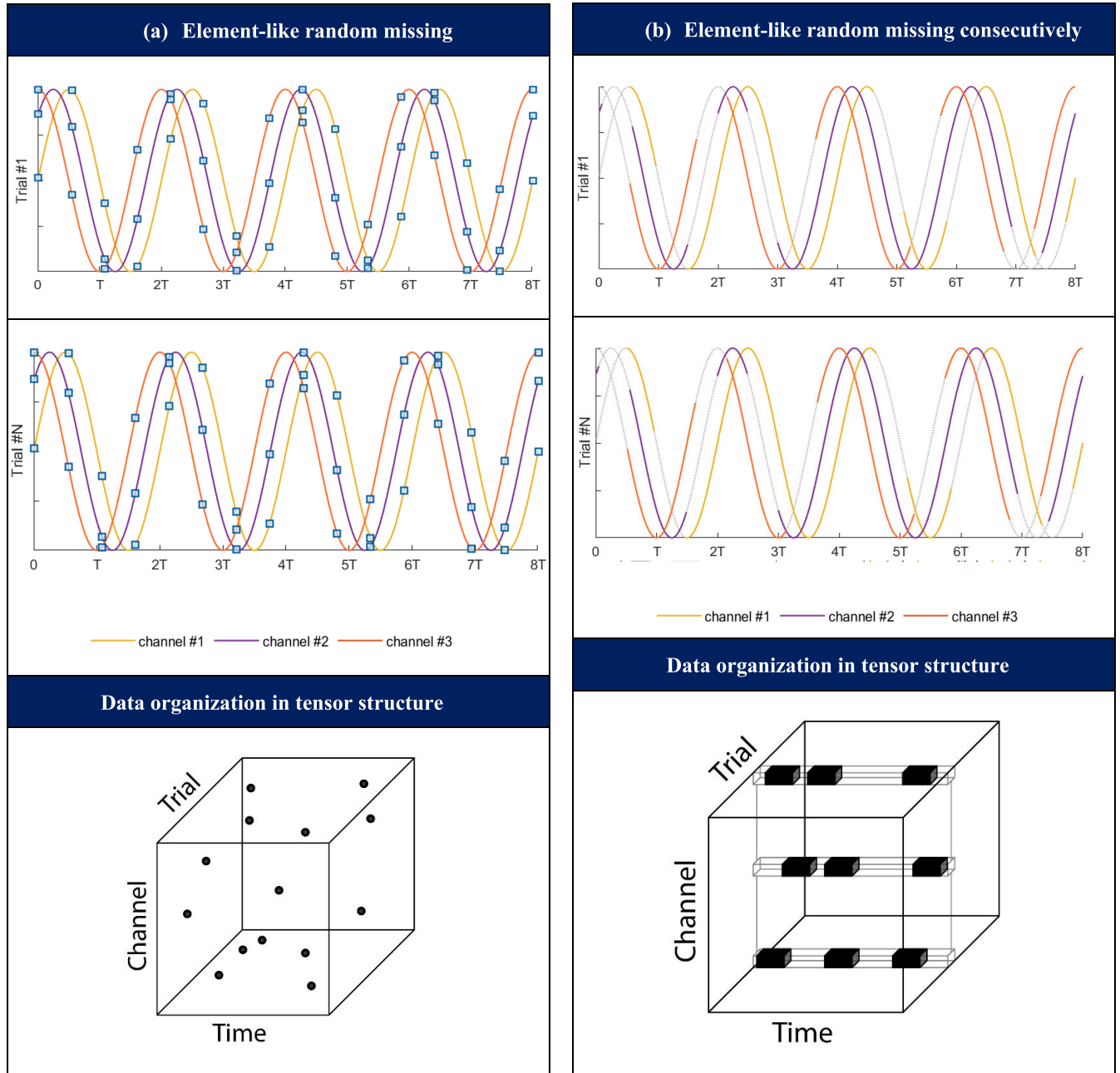


FIGURE 3. (a): An intuitive missing scenario of missing data patterns of EEG are given for element-like random missing (EM): small squares represent data that are missing at random. (b): An intuitive missing scenario of missing data patterns of EEG are given for consecutive element-like random missing (CEM): occlusion (segments in gray dot) denotes data that are missing consecutively.

where $\mathbf{X}_{it}$ is the actual value and $\hat{\mathbf{X}}_{it}$ is the corresponding imputed values, $\mathbf{W}$ is missing values indication matrix which is defined as follow:

$$\mathbf{W}_{it} = \begin{cases} 1 & \text{if } \mathbf{X}_{it} \text{ is an observed value} \\ 0 & \text{if } \mathbf{X}_{it} \text{ is a missing value} \end{cases}$$

Similarly, for a tensor-based imputation evaluation, the original time series tensors and the estimated time series tensors are unfolded into the vectorization form. Then, we apply (53) to calculate the RMSE between them.

Classification Accuracy: Accuracy assessment using confusion matrix is performed as an evaluation matrix for classification performance, as given:

$$\text{accuracy} = \frac{TP + TN}{TP + FP + FN + TN} \times 100$$

where TP and FN indicate the true and false diagnosis of motor imagery task of left hand while TN and FP denote the true and false diagnosis of imagination of movement of the right hand, respectively.

TABLE 2. Performance comparison of different imputation approaches for element-like random missing values on Gr, Fr-aw, Fr-al, Ri-SubA, Ri-SubB dataset, respectively. The average of RMSE varies by the growth of missing ratio.

Dataset/ Subject	Imputation Algorithm		Average RMSE with respect to percentage of missing values rate (%)				
			0.1	0.2	0.3	0.4	0.5
Gr	Interpolation	Linear	0.012791	0.025099	0.042692	0.063805	0.091871
		Spline	0.113456	0.145852	0.171667	0.189224	0.202246
	MSVD		0.077234	0.157395	0.231897	0.302367	0.370337
	Kalman Filter		0.022245	0.046686	0.075769	0.104916	0.138082
	Proposed method		0.006434	0.013145	0.022468	0.034216	0.054632
Fr	Interpolation	Linear	0.010296	0.020955	0.031492	0.043071	0.052716
		Spline	0.195512	0.688766	1.073513	1.235033	1.741851
	MSVD		0.168168	0.168903	0.169369	0.169975	0.170664
	Kalman Filter		0.003272	0.007017	0.01155	0.016736	0.022053
	Proposed method		0.003095	0.005933	0.008529	0.010989	0.013139
	Interpolation	Linear	0.002489	0.005131	0.007941	0.010768	0.013961
		Spline	0.304814	0.672311	1.047033	1.290092	1.978428
	MSVD		0.029132	0.029225	0.029342	0.029445	0.029569
	Kalman Filter		0.001518	0.002432	0.003672	0.005099	0.006708
	Proposed method		0.001133	0.002193	0.003107	0.003974	0.004803
Ri	Interpolation	Linear	0.019511	0.022491	0.025703	0.030439	0.034778
		Spline	0.152754	0.289654	0.374705	0.464873	0.489193
	MSVD		0.101245	0.184838	0.267324	0.430808	0.445231
	Kalman Filter		0.094142	0.177611	0.260714	0.324832	0.394775
	Proposed method		0.002407	0.006643	0.012632	0.022838	0.031827
	Interpolation	Linear	0.015793	0.028293	0.033074	0.041388	0.049782
		Spline	0.336109	0.674441	0.835318	1.073285	1.164548
	MSVD		0.023504	0.035312	0.055698	0.099954	0.111524
	Kalman Filter		0.013209	0.028821	0.042159	0.076882	0.089049
	iTa-DFiE		0.001652	0.003664	0.006425	0.014356	0.018652

TABLE 3. Performance comparison of different imputation approaches for consecutive element-like random missing values on Gr, Fr-aw, Fr-al, Ri-SubA, Ri-SubB dataset, respectively. Randomly 10% to 50% of original dataset with length of 50 are treated as “consecutive missing”. The average of RMSE varies by the growth of missing ratio.

Dataset/ Subject	Imputation Algorithm		Average RMSE with respect to percentage of missing values rate (%)				
			0.1	0.2	0.3	0.4	0.5
Gr	Interpolation	Linear	0.086991	0.181353	0.268778	0.307623	0.385849
		Spline	0.077618	0.150009	0.232712	0.307115	0.368631
	MSVD		0.151422	0.189946	0.271882	0.296584	0.352869
	Kalman Filter		0.082954	0.151076	0.232077	0.279667	0.339983
	Proposed method		0.024842	0.056034	0.088186	0.122012	0.173306
Fr	Interpolation	Linear	0.045455	0.083945	0.116912	0.155829	0.185067
		Spline	0.22236	0.792294	1.097909	1.781562	2.203046
	aw	MSVD	0.171764	0.175726	0.177438	0.186603	0.190132
		Kalman Filter	0.005085	0.009979	0.019541	0.031487	0.043805
	Proposed method		0.004824	0.008495	0.014179	0.020769	0.025498
	al	Linear	0.008437	0.017114	0.024701	0.030559	0.038018
		Spline	0.416131	0.808082	1.188171	2.058619	2.776966
		MSVD	0.029482	0.029879	0.030427	0.030989	0.031689
		Kalman Filter	0.001564	0.003061	0.005045	0.007286	0.010512
	Proposed method		0.001424	0.002737	0.004181	0.005326	0.006936
Ri	subA	Linear	0.100674	0.203122	0.288914	0.378263	0.468167
		Spline	0.13202	0.308931	0.367919	0.454175	0.556766
		MSVD	0.093603	0.190606	0.264361	0.371482	0.455914
		Kalman Filter	0.090473	0.183857	0.248268	0.342229	0.400887
	Proposed method		0.020978	0.023744	0.03232	0.050679	0.073932
	subB	Linear	0.110785	0.239865	0.311938	0.440843	0.522086
		Spline	0.277045	0.619573	0.883248	0.922155	1.092916
		MSVD	0.023439	0.033277	0.033546	0.060343	0.078011
		Kalman Filter	0.025403	0.4310430	0.068448	0.087438	0.095687
	Proposed method		0.004461	0.016881	0.023884	0.065635	0.087387

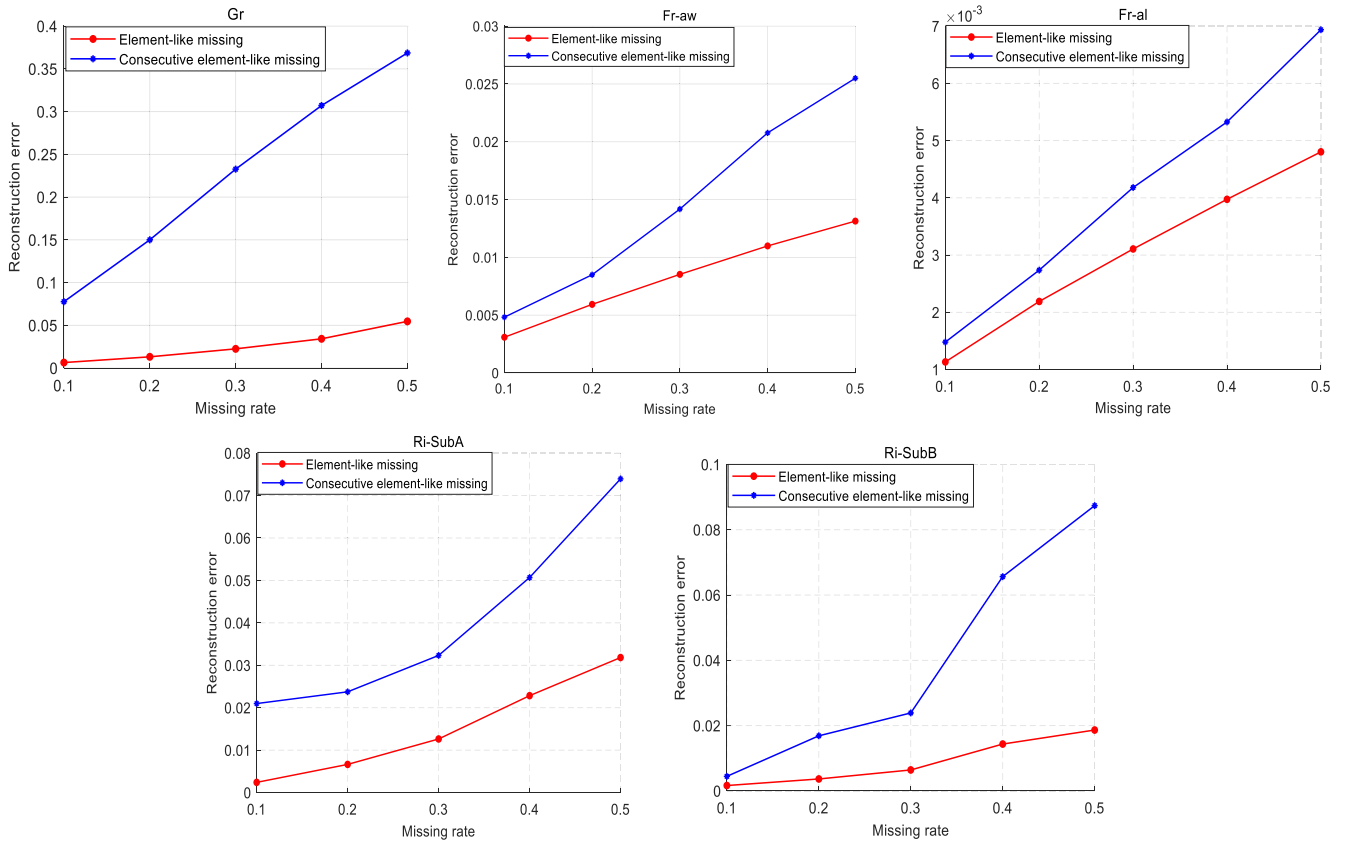


FIGURE 4. Proposed method performance comparison of element-like and consecutive element-like random missing data imputation over missing rates ranging from 10 % to 50% on Gr, Fr (Fr-aw and Fr-al) and Ri (Ri-SubA and Ri-SubB) dataset, respectively. Note that we fixed missing length of 50 time-ticks that treated as values missing continuously for the consecutive element-like random missing values scenario.

B. EXPERIMENT RESULTS AND DISCUSSION

1) TENSOR COMPLETION - RECONSTRUCTION ERROR

We performed the effectiveness of our proposed method on two missing scenarios, including EM and CEM as designed in Figure 3. For a fair comparison as well as to reduce random effect, we repeat each experiment of all algorithms 20 times and report the average of RMSE. By default, we set the dimension of latent factor $J = [3], [5], J = [10, 3]$ and $J = [5, 5]$ for Gr, Fr and Ri respectively in all the experiment evaluations.

Table 2 summarizes the completion performance of different imputation algorithms for EM scenario on the three EEG-based BCI dataset. The comparisons are conducted on different number of entries missing values over the ranging from 10% to 50%. As we can see, our proposed method performs very well for any missing rates with the lowest reconstruction error against all compared methods. Specifically, the performance of the linear interpolation imputation is slightly worse than that of the proposed method, ranking second among these algorithms. Original KF is slightly better than MSVD and spline completion, but its demonstration is worse than linear interpolation. MSVD and spline produce almost the worst results.

With similar experiments, the effectiveness of our proposed approach is again verified for CEM where occlusions of typical EEG-based BCI system often occur consecutively. To generate synthetic occlusions or known as consecutive missing values, we start at specific frame and setting a marker for the occlusion of this marker, then we set occlusion with 50 duration length as via using Poisson distributed random variable w.r.t the observed statistics, and repeating until obtaining occluded 10%, 20%, 30%, 40%, 50% of the original tensor input values, respectively. In all the datasets, we consider only one case of missing length as mentioned above because in real dataset of Gr just recording 350-time frame. The performance of missing data recovery is summarized in Table 3 with the same conclusions as EM cases, except that the MSVD imputation is better than interpolation for such long-term missing values.

Overall, the proposed method overwhelmingly outperforms the other two classical models (spline, linear interpolation); straightforward alternative (MSVD) and the original dynamics KF imputation method in diverse missing scenarios (EM and CEM scenarios with vary missing rates).

From the above perspectives, we can conclude that our proposed method provides a promising solution on the

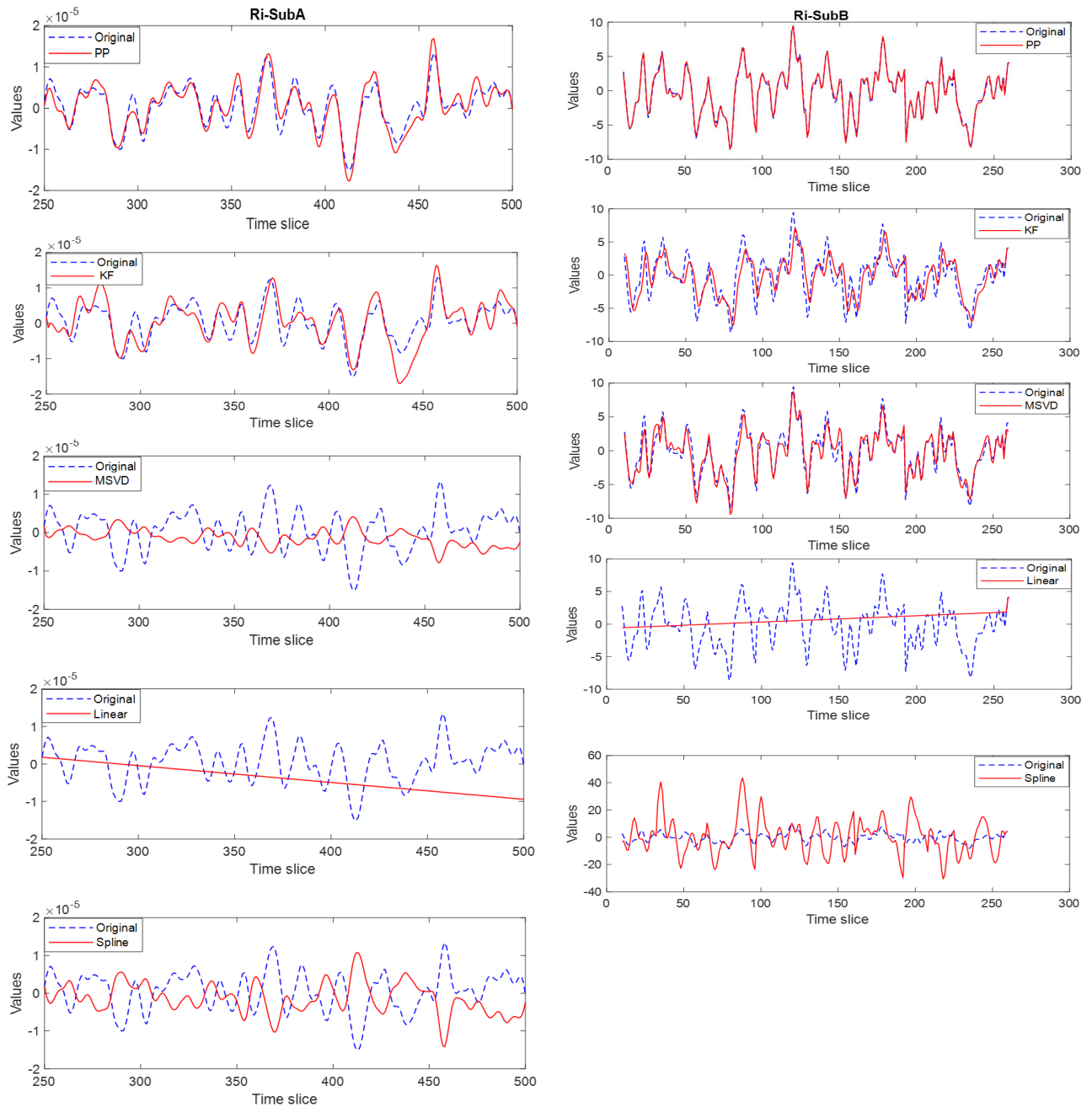


FIGURE 5. An illustration of estimation performances of different imputation algorithms for 250 time slices occlusion on Ri-SubA and Ri-SubB, respectively. The proposed method outperforms all competitors which get very close to the original. The estimated value and corresponding actual values of EEG data are under 10% missing rate. Top to bottom figures correspond to original signal vs reconstructed from PP, KF, MSVD, linear interpolation and splines respectively.

performance of missing data imputation under different missing rates in both EM and CEM time series. It means that the proposed method successfully incorporates brain dynamics, correlation among channels and tensor structure of the data. In the aspect of different missing scenarios, the experimental results demonstrate that CEM is more difficult to recover than the EM, shown in Fig. 4. As can be seen from Fig. 4, the reconstruction error RMSE of the imputation results of EM

patterns is lower than that of the CEM over different missing rates ranging from 10 % to 50%.

As expected, the proposed method has matched with a quantitative point of view on theoretical and algorithmic level since its demonstration on Fr dataset yields the best performance among the four datasets. This is due to the strong temporal smoothness and correlations in dataset Fr plays an important role in recovering the missing values.

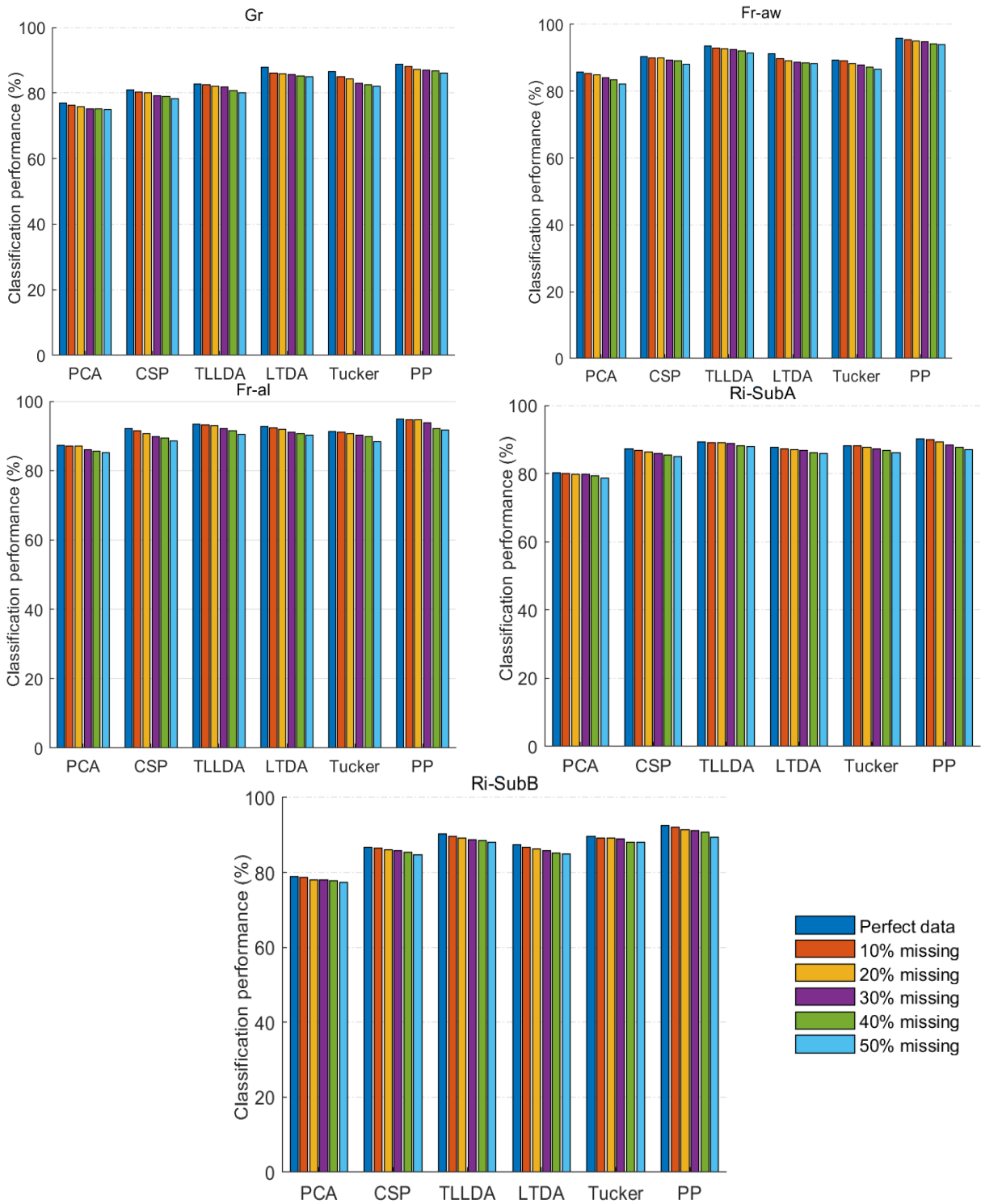


FIGURE 6. Classification performance is based on the extracted features of different decomposition methods with different missing rates. The proposed method demonstrates the best accuracy against others.

To make our approach more persuasive, two intuitive visualizations for comparing the recovery performance of our proposed approach with the other classical alternatives including spline, linear interpolation, MSVD and KF are illustrated in Fig. 5.

The experiment setting only shows for the scenario of consecutive element-like random missing with 10% the missing

rate of original tensor since this problem is more common and realistic in EEG missing patterns.

In detail, we implement by fixing 10% of missing rate and run different imputation algorithms for 250 time slices occlusion on Ri-SubA and Ri-SubB, respectively. As can be explicitly seen from each subfigure, our proposed method is visually superior to other competitors since the reconstructed

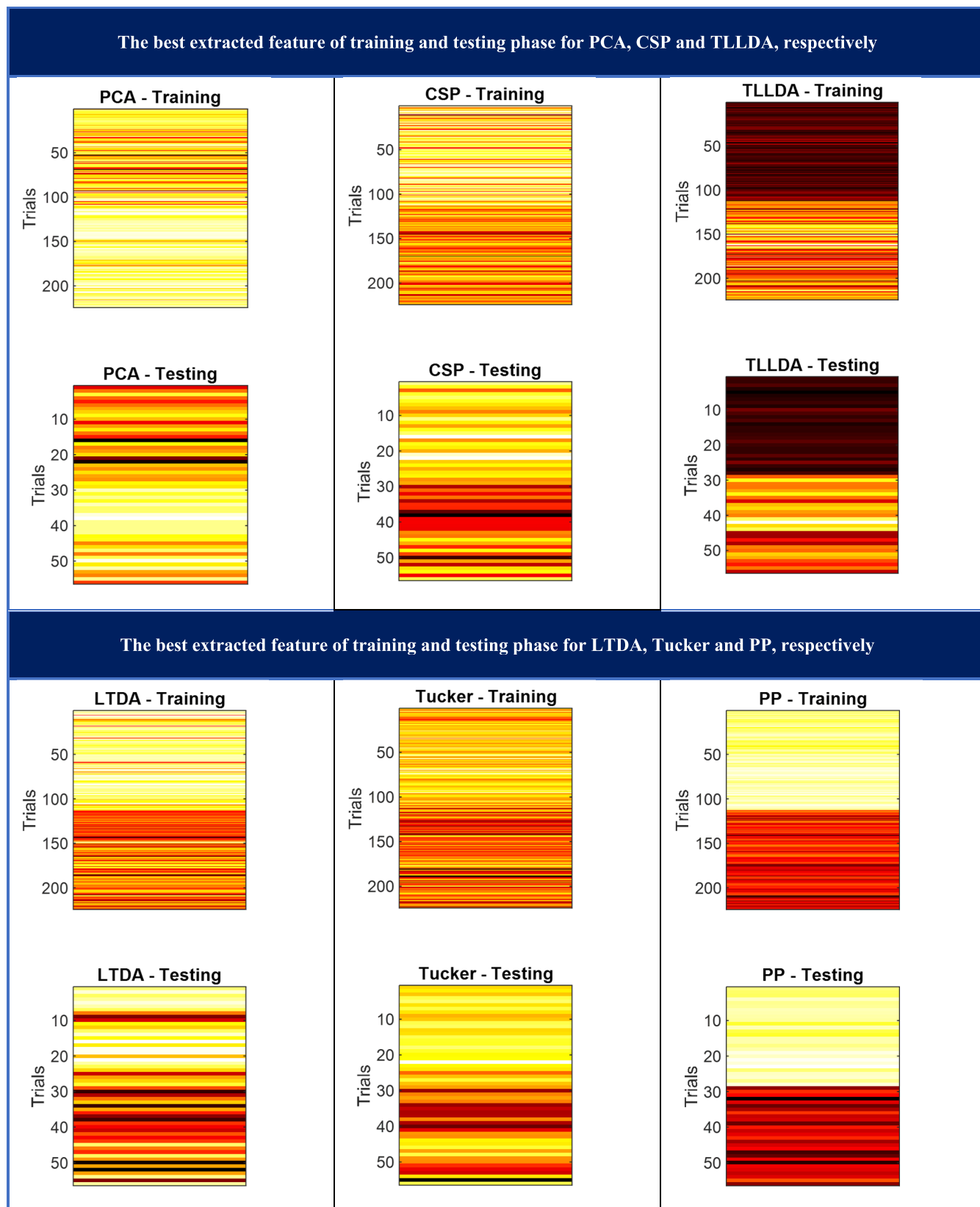


FIGURE 7. Visualization of the best extracted features of training and testing phases on Gr dataset of different features decomposition algorithms.

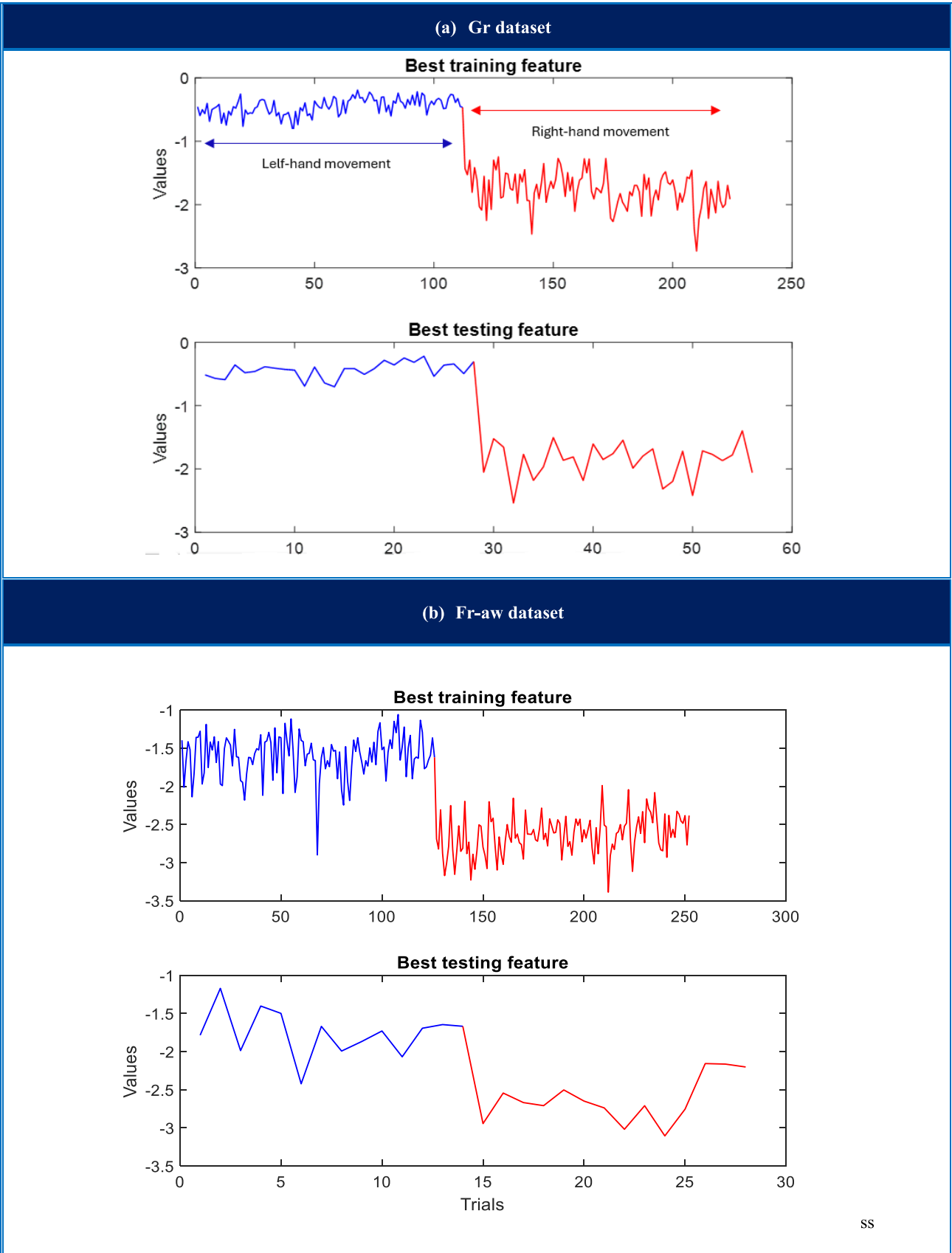
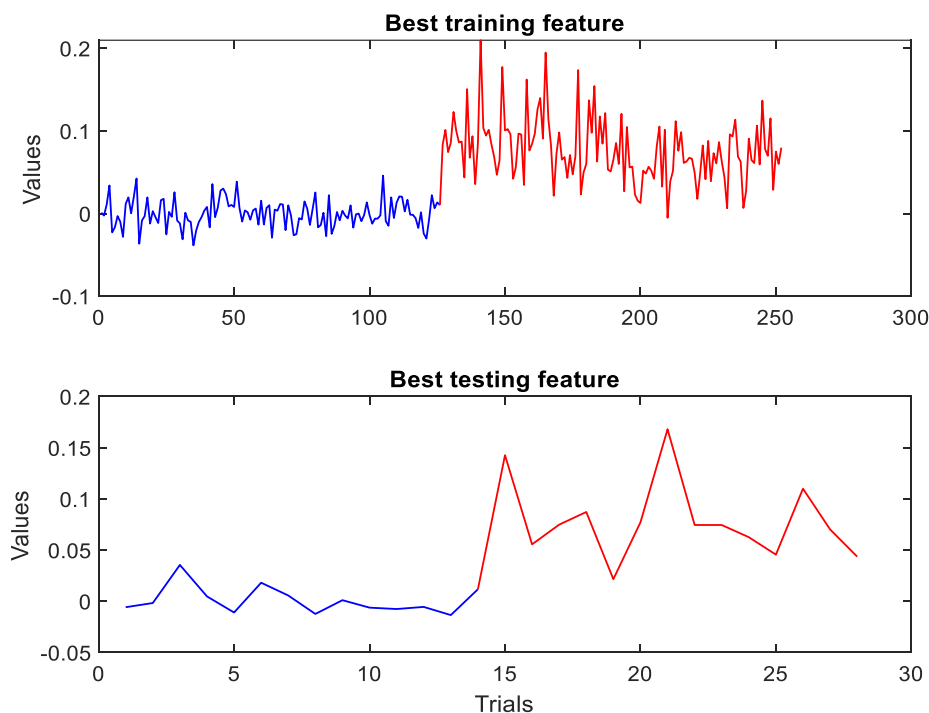


FIGURE 8. The best extracted features of proposed method for training and testing phase on Gr, Fr-aw, Fr-al, Ri-SubA and Ri-subB dataset, respectively.

(c) Fr-al dataset



(d) Ri-SubA dataset

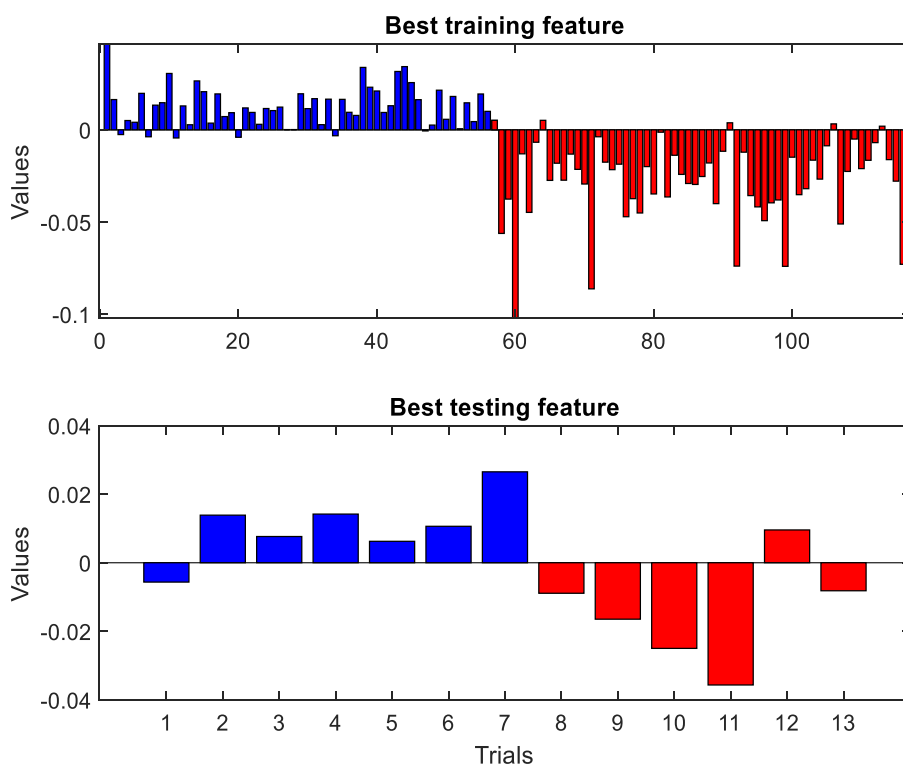


FIGURE 8. (Continued.) The best extracted features of proposed method for training and testing phase on Gr, Fr-aw, Fr-al, Ri-SubA and Ri-subB dataset, respectively.

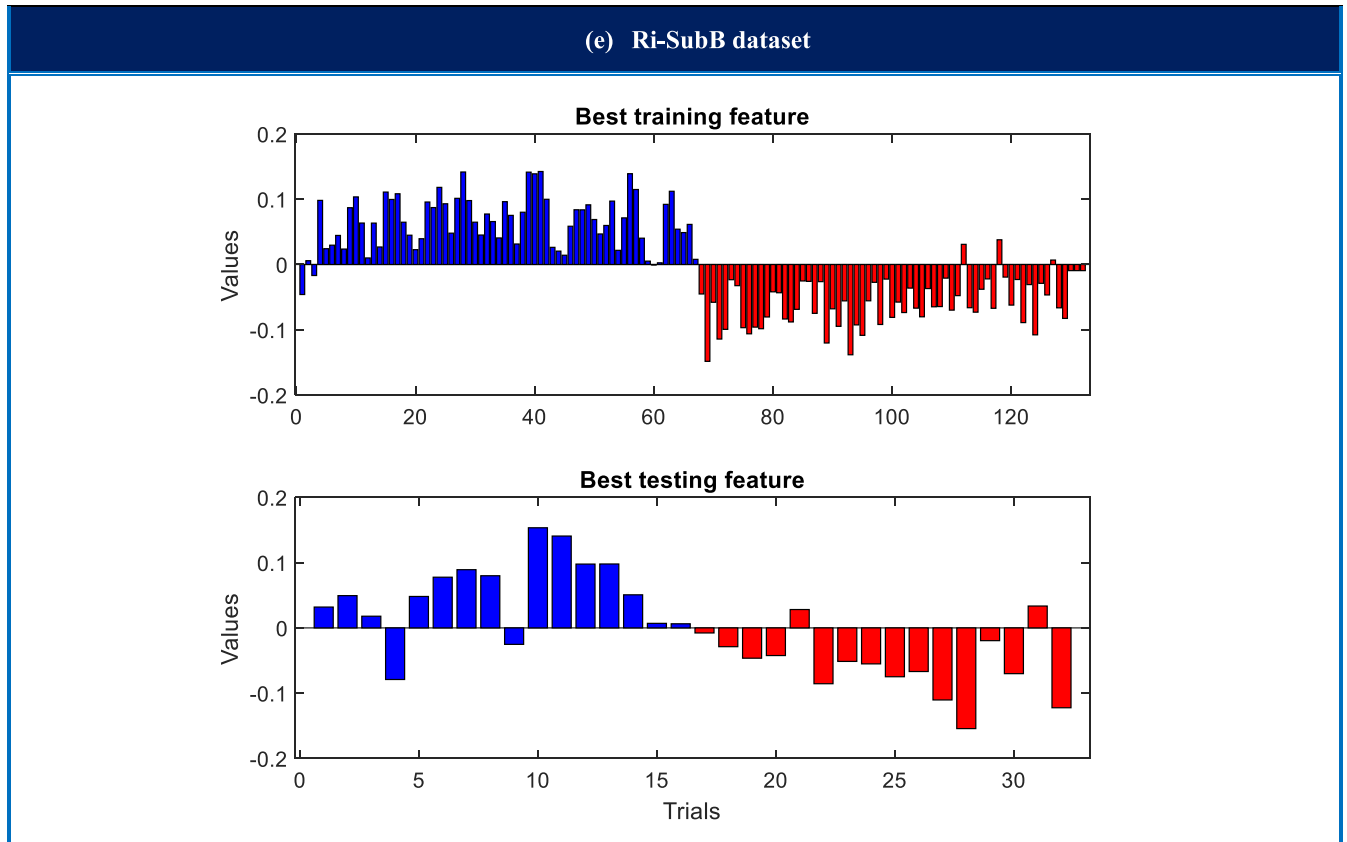


FIGURE 8. (Continued.) The best extracted features of proposed method for training and testing phase on Gr, Fr-aw, Fr-al, Ri-SubA and Ri-subB dataset, respectively.

signal does not deviate much from the original data. Obviously, the value imputed by the proposed method is nearly fitting to the corresponding real values. This proves that the proposed tensor completion works well than matrix-based completion approaches.

2) TENSOR DECOMPOSITION AND RECOGNITION

Once missing values are recovered using the proposed method via improving KF approach, feature extraction is then performed to boost the accurate classification for motor imagery tasks (imagery left or right-hand movements) of multichannel time-frequency of the EEG-based BCI. As such, the EEG data for each trial is formed as a 3-D tensor representation $\underline{\mathbf{X}} \in \mathbb{R}^{I_1 \times I_2 \times I_3}$ with modes channel, frequency bins and time frames, respectively. Consequently, the training dataset is formed a 4-D tensor of channel $\times$ frequency bins $\times$ time frames 224, 252, 252, 117, 130 trials for Gr, Fr-aw, Fr-al, Ri-SubA, and Ri-SubB dataset respectively. Similarly, the testing set is organized 3-D sub-tensors of the remaining number trials of each dataset respectively. In our implementation, the time temporal domain is transferred into time-frequency domain by applying the complex Morlet wavelets transformation for all EEG signals.

The proposed tensor decomposition is then applied directly on the origin EEG tensor structure in order to extract the

meaningful features via finding three discriminant basic factors $\mathbf{A}^{(1)}, \mathbf{A}^{(2)}, \mathbf{A}^{(3)}$ which corresponds to spatial, spectral and temporal component, respectively. In our system, the threshold of energy criterion to determine number of factors in each mode is set up to 95%. The classification is carried out with a hold/out ratio of 80-20% mechanism with selected significant features. We select the CEM scenarios with 10% missing rates and 50 occlusions in length to evaluate the feasibility of the proposed method. The SVM classifier using the Gaussian Radial Basis Function kernel was used. The performance results are implemented over 10 runs with average accuracy shown in Fig. 6.

As can be seen from the bar chart, the proposed method outperforms other methods in all datasets (both on matrix-based and tensor-based decomposition). Taken Gr dataset as an illustration, the average performance obtained by proposed method on complete Gr dataset is 88.75% which was improved by approximately 15.45%, 9.63%, 7.19%, 0.92%, 2.56% compared with PCA, CSP, TLLDA, LTDA and Tucker, respectively. Fig. 6 shows the proposed method still performs the best accuracy compared with others for varying missing rates. Specifically, the classification result on 50% missing rate of CEM scenarios obtained 86.12% accuracy which was decrease 2.9% compared with performance on the complete dataset. While other methods including PCA, CSP,

TLLDA, TLDA, Tucker on this such type come down 2.5%, 3.3%, 3.1%, 3.2%, 5.1% compared with on the perfect data, respectively.

Overall, the best performance for classification is achieved on complete dataset for all datasets. As expected, the proposed tensor decomposition convinces its effectiveness with the highest accuracy on these complete datasets compared with others. Moreover, the proposed approach provides similar accuracies for incomplete datasets but slightly decreases when the missingness ratio increases. This proves that only the proposed method can exploit the hidden factors successfully that lead to revealing the discriminative information between two classes to boost classification performance as well as visualization.

In order to deeply look into the compact and interpretable features between the proposed methods and others, taking the Gr dataset as an example, the heat-map of Fig. 7 shows the visualization of the best features of all dimensional decomposition methods. Each row of the hot-scaled heat-map indicates the trials while column corresponds to the best extracted features.

For additional confirmation of the effective features of the proposed tensor decomposition method on different datasets, Fig. 8 shows the best features on training and testing phase for Gr, Fr-aw, Fr-al, Ri-SubA and Ri-SubB, respectively. Intuitively, our proposed method gives the expected values that clearly identify the left-hand movement and right-hand movement tasks that display in blue and red color, respectively.

Similarly, get back to deeply analyzing in Fig. 7, it is highlighted that the method yields similar rows color visually means that they have the same classes and thus not supervising since the proposed method obtained the highest classification accuracy. While the typical matrixed-based method for dimensionality reduction PCA gives the worst results since the rows do not clearly distinguish each other. Three tensor-based methods, TLLDA, LTDA and Tucker are more favorable than those of matrixed-based including PCA and CSP.

VI. CONCLUSION

Missing brain signals data imputation is a crucial task of data quality enhancement in the data driven intelligent BCI systems. The paper is interested with respect to scientific modeling of such brain activity, specifically in recognition for incomplete multimodal EEG-based BCI data involving left/right motor-imagination movements. The main objective is to propose and design a new paradigm, iTa-DFiE, on a theoretical and an algorithmic level relying on the recent advances in tensor algebra without breaking the original tensor structure of the EEG dataset. iTa-DFiE demonstrates its original and utility via proposing new approach for tensor completion and tensor decomposition via refinement of the original of Kalman Filter model and mathematical framework of TUCKER decomposition, respectively. The recognition performances on three real datasets demonstrated the effectiveness of our proposed framework that overwhelmingly

outperforms compared with the others on missing values recovery as well as feature extraction tasks. Expectedly, the valuable outcomes of this article will be helpful to extend and develop an effective decision-making system for high-order incomplete dataset on other real-life applications in time series domains.

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NGOC ANH THI NGUYEN received the B.S. degree from the Faculty of Mathematics and Informatics, The University of Danang–University of Science and Education, Vietnam, in 2006, and the M.S. and Ph.D. degrees from the Department of Electronics and Computer Engineering, Chonnam National University, Gwangju, South Korea, in 2011 and 2016, respectively. She is currently a Lecturer with The University of Danang–University of Science and Education. Her research interests include data mining, machine learning, big data analysis, artificial intelligence, pattern recognition, mathematical modeling, and their applications in bioinformatics and neuroscience.



data-driven machine learning.

QUANG-BANG TAO received the B.S. degree from The University of Danang–University of Science and Technology, Vietnam, the M.S. degree from Chonnam National University, South Korea, and the Ph.D. degree from Paris-Saclay University, France, in 2016. He is currently a Professor with the Department of Mechanical Engineering, The University of Danang–University of Science and Technology. His current research interests include computational engineering and



His research interests include database and database management systems, data mining, software engineering, and science education.

NGUYEN TRAN QUOC VINH received the B.Sc. degree in computer science, the M.Sc. degree in system software, and the Ph.D. degree in computerized control automated systems and progressive information technology from Odessa National Polytechnic University, Ukraine, in 2001, 2002, and 2006, respectively. From 2007 to 2013, he was with the University of Da Nang–University of Economics. He is currently a Lecturer with the Faculty of Information Technology, The University of Danang–University of Science and Education.



HYUNG-JEONG YANG (Member, IEEE) received the B.S., M.S., and Ph.D. degrees from Chonbuk National University, South Korea. She is currently a Professor with the Department of Artificial Intelligence Convergence, Chonnam National University. Her main research interests include multimedia data mining, medical data analysis, social network service data mining, and video data understanding.



His research interests include computational intelligence for image analysis, pattern recognition, and medical data analysis.

HIEU TRUNG HUYNH (Member, IEEE) received the B.S. and M.S. degrees in computer engineering from Ho Chi Minh City University of Technology, Vietnam, in 1998 and 2003, respectively, and the Ph.D. degree in computer engineering from Chonnam National University, South Korea, in 2009. From 1998 to 2010, he was with Ho Chi Minh City University of Technology. He was a Visiting Scholar with Osaka Sangyo University, Japan, from 2001 to 2002, a Postdoctoral Researcher with



Technology. His research interests include signal processing, data science, and machine learning.

DUY KHANH NINH received the B.S. degree in electronics and telecommunication technology from Vietnam National University, Hanoi, in 2003, the M.S. degree in information processing and communications from Hanoi University of Science and Technology, in 2006, and the Ph.D. degree in information science from Ritsumeikan University, in 2016. Since 2006, he has been a Senior Lecturer with the Faculty of Information Technology, The University of Danang–University of Science and

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